Multinomial Credits Analysis

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Load **survey_data:**

- survey responses from qualtrics
- includes: *5bi) How many credits of instruction in quantitative Study Design do students receive in core (compulsory) teaching across the entire undergraduate degree?* and *5ci) How many credits of undergraduate instruction in Data Analysis do students receive in core (compulsory) teaching across the entire undergraduate degree?*

```
#load survey data
survey_data <- read_rds(here("data/clean_data", "clean_survey_data.rds"))

#look at data
survey_data %>%
  select(university_id, discipline, credits_sd, credits_da) %>%
  #sd = study design
  #da = data analysis
  glimpse()
```

```
## Rows: 103
## Columns: 4
## $ university_id <fct> u90, u90, u67, u67, u66, u67, u97, u86, u93, u12, u87, N~
## $ discipline    <fct> Bio, Psych, Bio, Med, Bio, Psych, Biomed, Biomed, Med, B~
## $ credits_sd    <ord> 20-29, 30-39, 20-29, 10-19, 1-9, 20-29, 20-29, 1-9, 1-9,~
## $ credits_da    <ord> 60-69, 40-49, 10-19, 1-9, 20-29, 50-59, 20-29, 10-19, 1-~
```

```
#table: study design credits by discipline
table(survey_data$discipline, survey_data$credits_sd)
```

```
##
##           0 1-9 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80+
##   Bio     0  12     5     8     2     2     1     0     0   1
##   Biomed  3   7     4     3     0     1     0     0     0   1
##   Med     0   5     6     1     1     1     1     0     0   0
##   Psych   0   1     4    11     3     4     1     6     0   8
```

```
#table: data analysis credits by discipline
table(survey_data$discipline, survey_data$credits_da)
```

```
##
##           0 1-9 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80+
##   Bio     0   6     9     8     2     3     1     1     0   1
##   Biomed  0   2    10     5     0     0     0     0     0   2
##   Med     0   8     2     4     0     0     1     0     0   0
##   Psych   0   1     1     4     8     7     2     7     0   8
```

Credits: Study Design

Models with 4 categories

recode categories

Recoding from 10 categories to 4 reduces the number of response categories with zero responses.

```
#recode categories
survey_data <-
  survey_data %>%
  mutate(
    credits_sd_recoded = fct_collapse(credits_sd,
                                      "0-19" = c("0", "1-9", "10-19"),
                                      "20-39"= c("20-29","30-39"),
                                      "40-59" = c("40-49", "50-59"),
                                      "60+" = c("60-69", "70-79", "80+")))

#structure: ordered factor with 4 levels
glimpse(survey_data$credits_sd_recoded)
```

```
##  Ord.factor w/ 4 levels "0-19"<"20-39"<..: 2 2 2 1 1 2 2 1 1 1 ...
```

```
#table: study design credits (4 categories) by discipline
table(survey_data$discipline, survey_data$credits_sd_recoded)
```

```
##
##          0-19 20-39 40-59 60+
##   Bio      17    10     3   1
##   Biomed   14     3     1   1
##   Med      11     2     2   0
##   Psych     5    14     5  14
```

Multinomial Analysis of Study Design Credits

We now fit a multinomial model, using a prior inspired by Gelman et al. (2008) (we use the recommended values for standard deviation, but use normal distributions rather than cauchy).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```
#We obtain the default prior, and then update the prior, below.
get_prior(credits_sd_recoded ~ discipline, family = categorical(link = "logit"), data = survey_data)
```

```
##                 prior     class             coef group resp   dpar nlpar lb ub
##                (flat)         b                                mu2039
##                (flat)         b disciplineBiomed             mu2039
##                (flat)         b    disciplineMed             mu2039
##                (flat)         b  disciplinePsych             mu2039
##  student_t(3, 0, 2.5) Intercept                                mu2039
##                (flat)         b                                mu4059
##                (flat)         b disciplineBiomed             mu4059
##                (flat)         b    disciplineMed             mu4059
##                (flat)         b  disciplinePsych             mu4059
##  student_t(3, 0, 2.5) Intercept                                mu4059
##                (flat)         b                                  mu60
##                (flat)         b disciplineBiomed               mu60
##                (flat)         b    disciplineMed               mu60
```

```
##                 (flat)         b  disciplinePsych              mu60
##  student_t(3, 0, 2.5) Intercept                                  mu60
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
```

```
#Our prior, following Gelman et al. (2008), but we use a normal (not cauchy) distribution:
priors.studydesign <- c(
  prior(normal(0, 10), class = "Intercept", dpar = "mu2039"),
  prior(normal(0, 10), class = "Intercept", dpar = "mu4059"),
  prior(normal(0, 10), class = "Intercept", dpar = "mu60"),
  prior(normal(0, 2.5), class = "b", dpar = "mu2039"),
  prior(normal(0, 2.5), class = "b", dpar = "mu4059"),
  prior(normal(0, 2.5), class = "b", dpar = "mu60")
)

contrasts(survey_data$discipline) =contr.sum(4)
contrasts(survey_data$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

```
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1

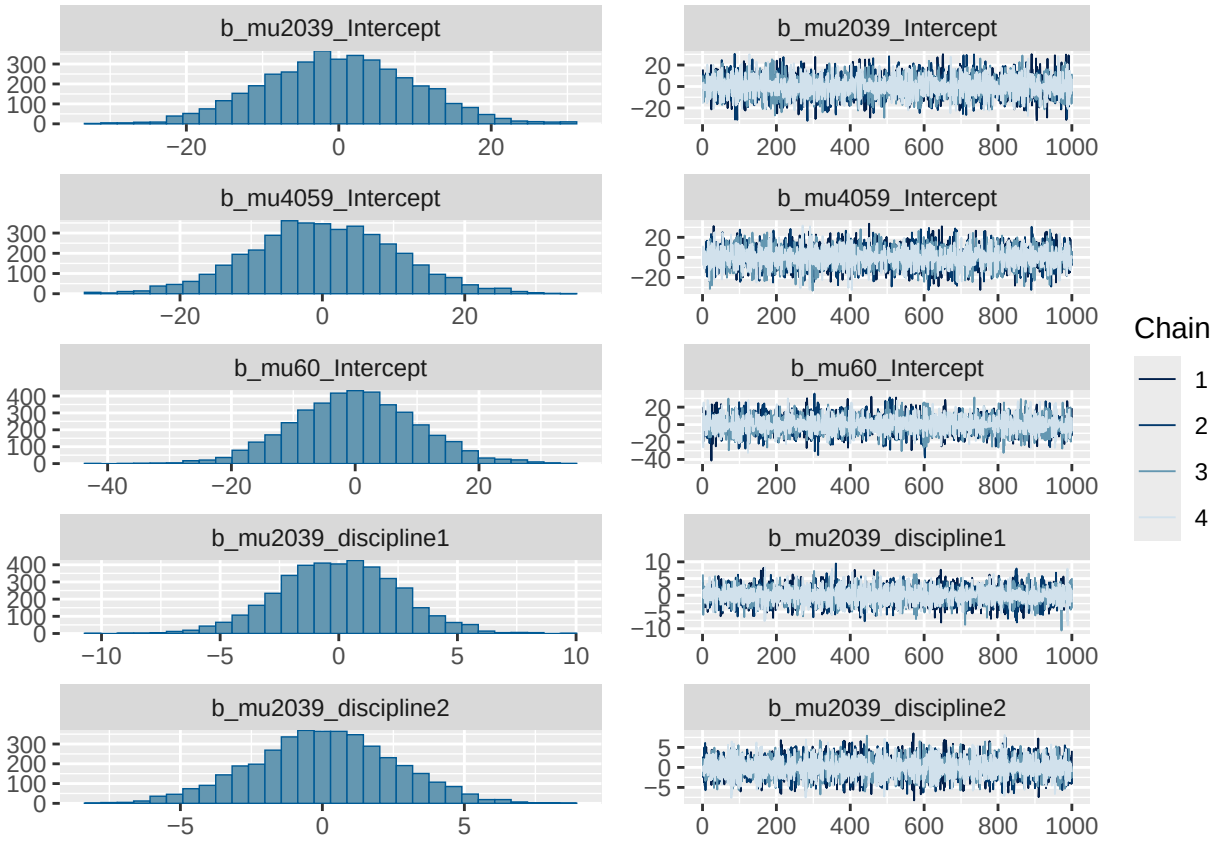

#Prior predictive check:
brm.formula1.sd <- brmsformula(credits_sd_recoded ~ discipline,
                               family = categorical(link = "logit"))

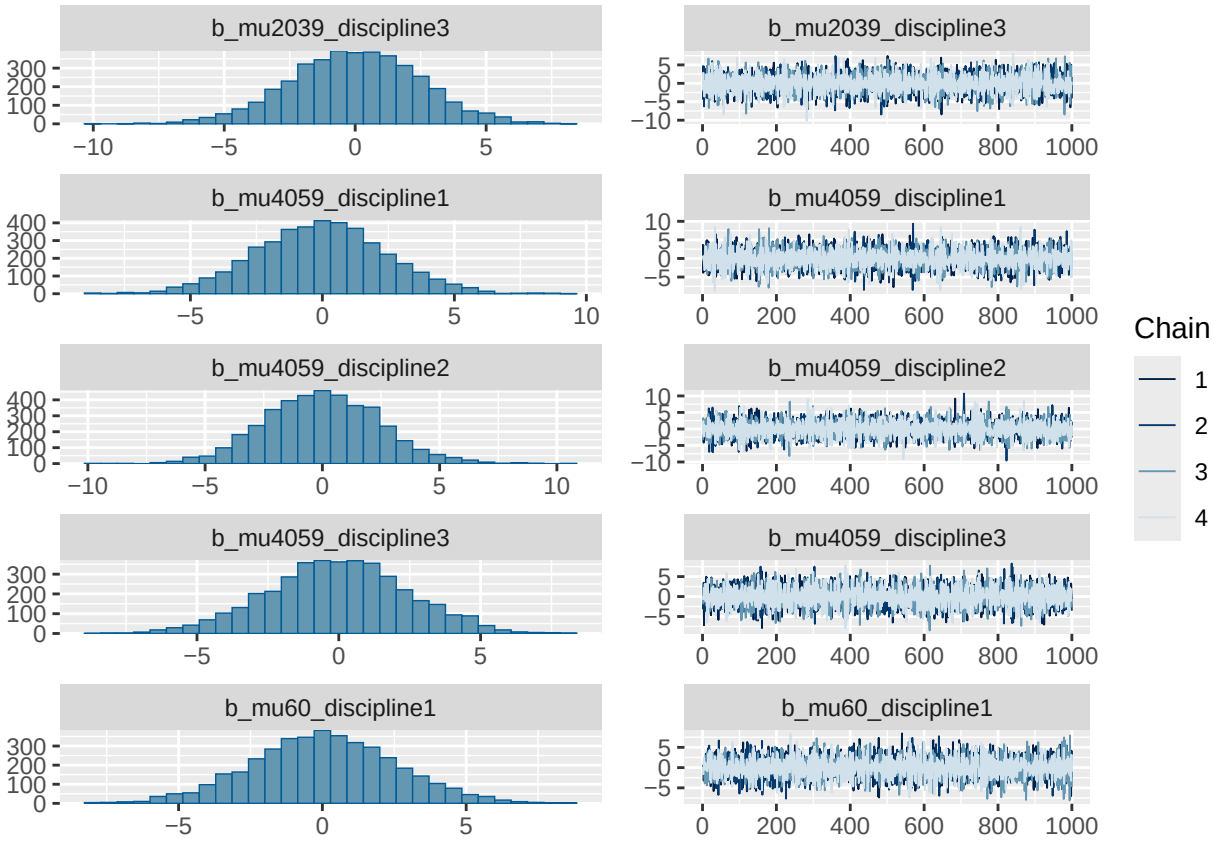
brm.mn.studydesign.priorpc <- brm(
  brm.formula1.sd,
  prior = priors.studydesign,
  sample_prior = "only",
  data = survey_data,
  file = here("store/lists", "brm.mn.studydesign.priorpc"))
```

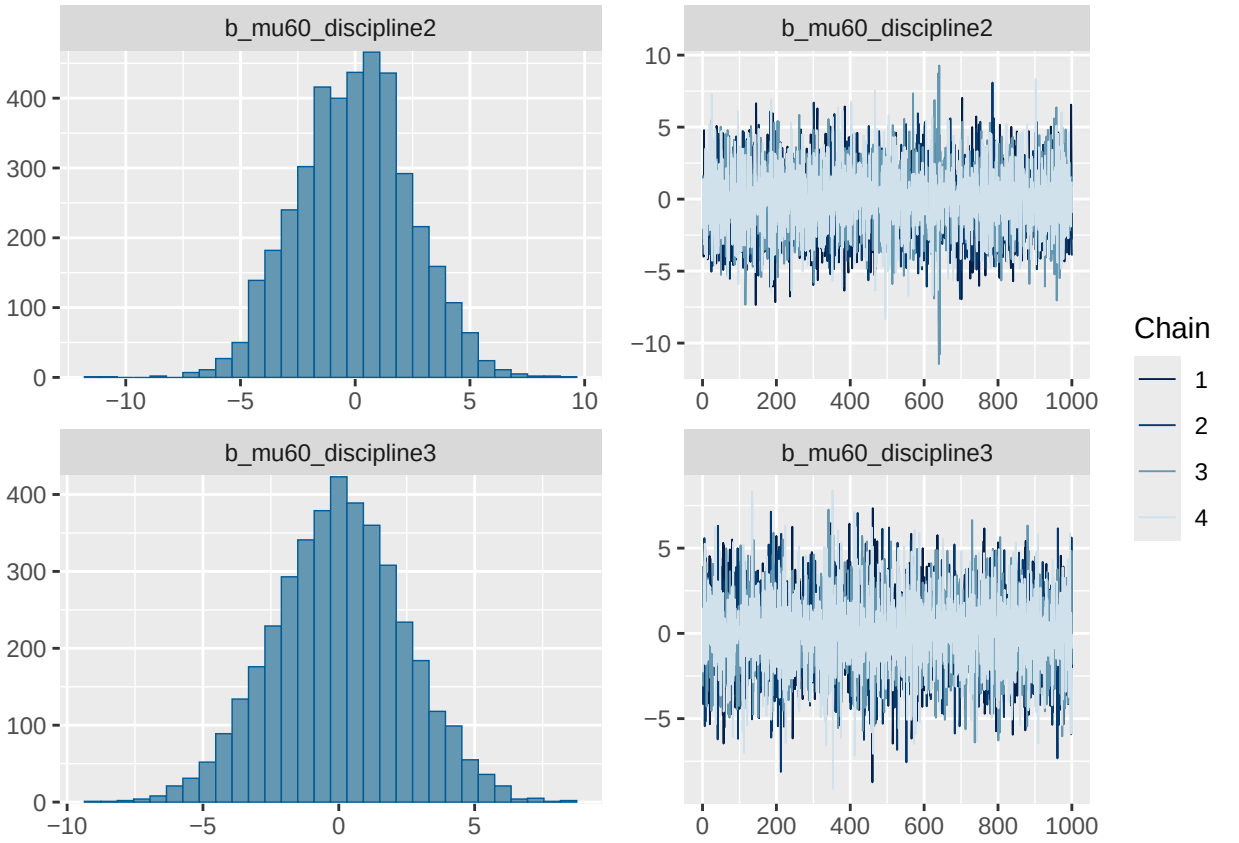
```
summary(brm.mn.studydesign.priorpc)
```

```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_sd_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept        0.05      9.83   -18.85    19.18 1.00     8244     3327
## mu4059_Intercept        0.11     10.29   -20.26    20.36 1.00     8371     2604
## mu60_Intercept         -0.04     10.06   -19.32    19.69 1.00     8417     2577
## mu2039_discipline1     -0.01      2.51    -5.00     4.99 1.00     9615     2791
## mu2039_discipline2      0.06      2.51    -4.94     4.90 1.00     9205     2872
## mu2039_discipline3     -0.01      2.47    -5.03     4.85 1.00     8782     3292
## mu4059_discipline1     -0.06      2.46    -4.92     4.79 1.00     9307     2741
## mu4059_discipline2     -0.01      2.44    -4.70     4.92 1.00     8106     2739
## mu4059_discipline3     -0.03      2.47    -4.94     4.73 1.00     9285     3038
## mu60_discipline1        0.05      2.51    -4.97     5.00 1.00    10370     2651
## mu60_discipline2       -0.05      2.46    -4.63     4.70 1.00    10248     3074
## mu60_discipline3        0.02      2.41    -4.73     4.79 1.00     7908     3318
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

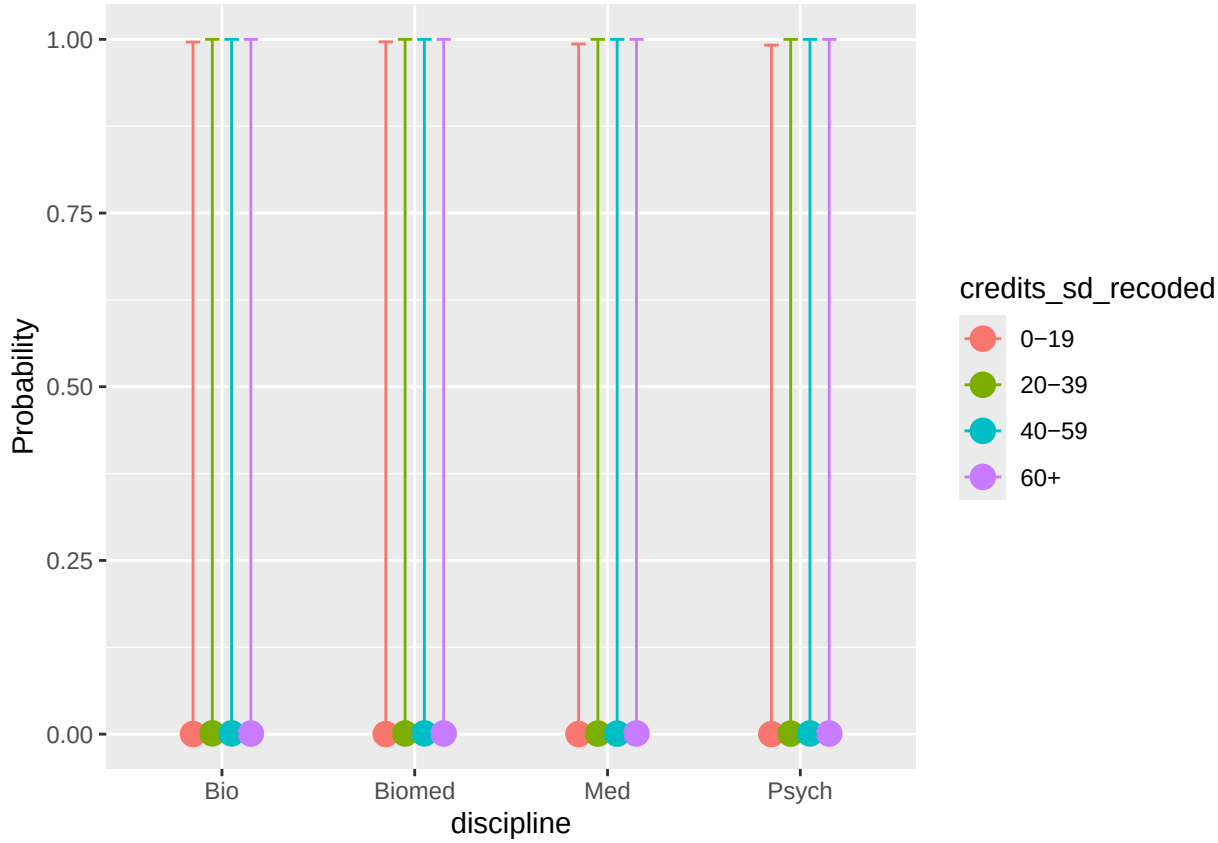
```
plot(brm.mn.studydesign.priorpc)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
Chain
1
2
3
4


b_mu2039_discipline3
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline2
b_mu4059_discipline3
b_mu4059_discipline3
b_mu60_discipline1
b_mu60_discipline1
Chain
1
2
3
4




```
#Prior predictive check outout:
conditional_effects(brm.mn.studydesign.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.studydesign.priorpc)` and `plot(brm.mn.studydesign.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman prior:
brm.mn.studydesign <- brm(credits_sd_recoded ~ discipline,
                          family = categorical(link = "logit"),
                          prior = priors.studydesign,
                          data = survey_data,
                          file = here("store/models", "brm.mn.studydesign"))

#Examine prior that we specified:
prior_summary(brm.mn.studydesign)
```

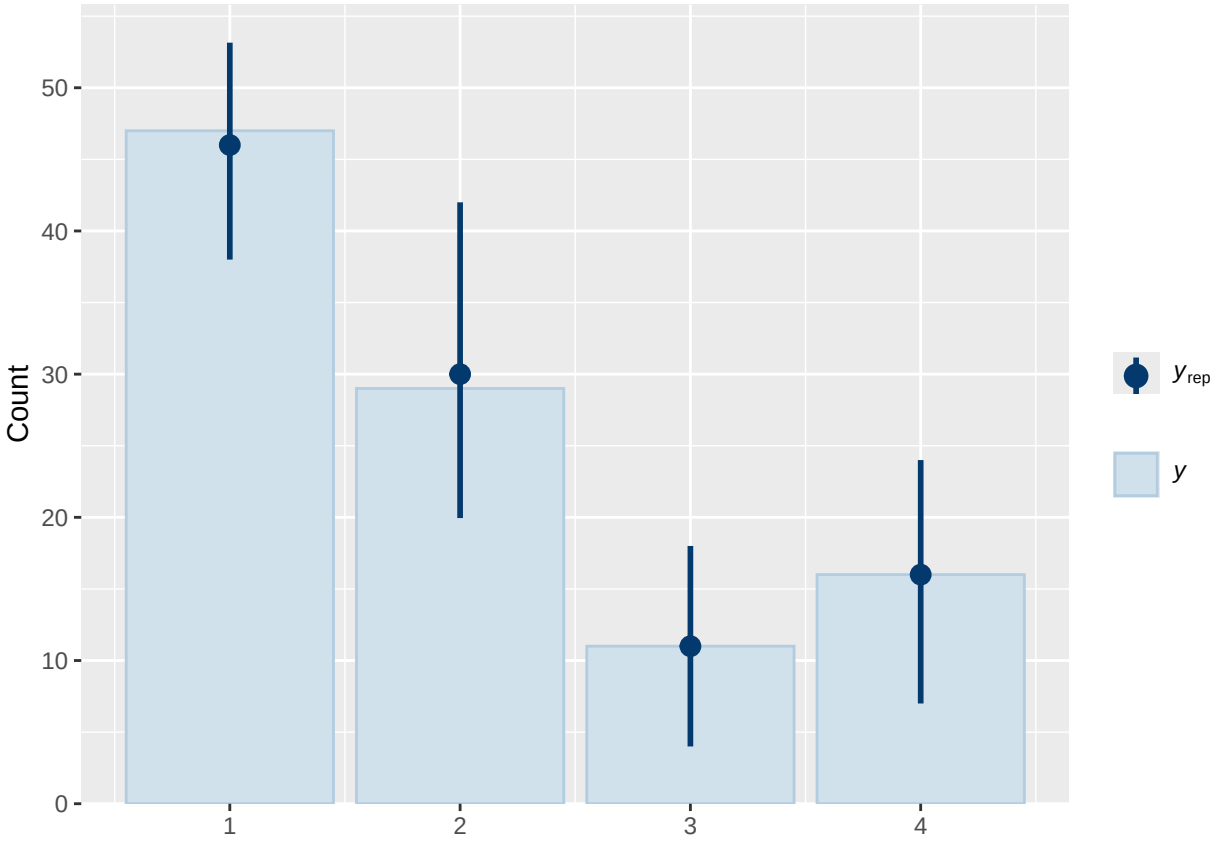
```
##           prior     class        coef group resp   dpar nlpar lb ub
##  normal(0, 2.5)         b                        mu2039
##  normal(0, 2.5)         b disciplines1           mu2039
##  normal(0, 2.5)         b disciplines2           mu2039
##  normal(0, 2.5)         b disciplines3           mu2039
##  normal(0, 2.5)         b                        mu4059
##  normal(0, 2.5)         b disciplines1           mu4059
##  normal(0, 2.5)         b disciplines2           mu4059
##  normal(0, 2.5)         b disciplines3           mu4059
##  normal(0, 2.5)         b                          mu60
##  normal(0, 2.5)         b disciplines1             mu60
##  normal(0, 2.5)         b disciplines2             mu60
```

```
##  normal(0, 2.5)         b discipline3              mu60
##   normal(0, 10) Intercept                         mu2039
##   normal(0, 10) Intercept                         mu4059
##   normal(0, 10) Intercept                           mu60
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
##          user
```

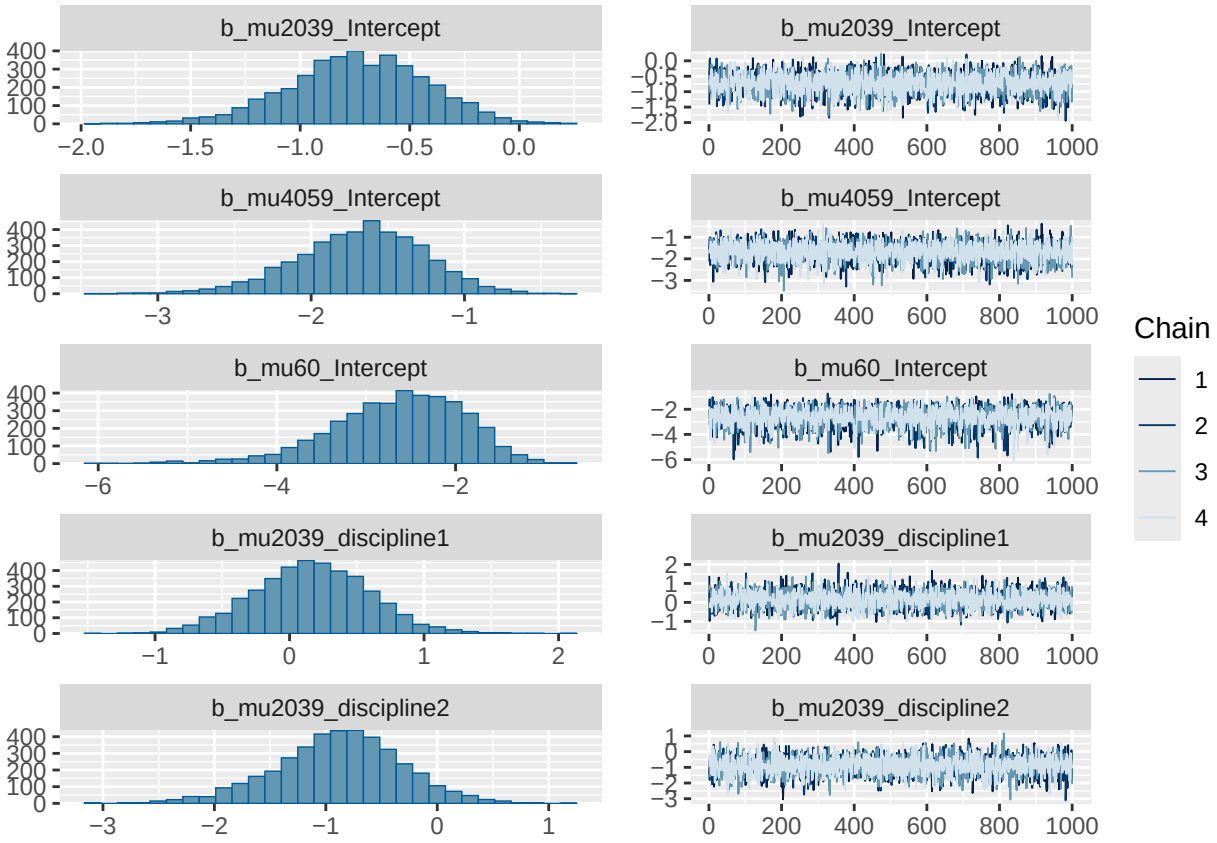
```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.studydesign)
```

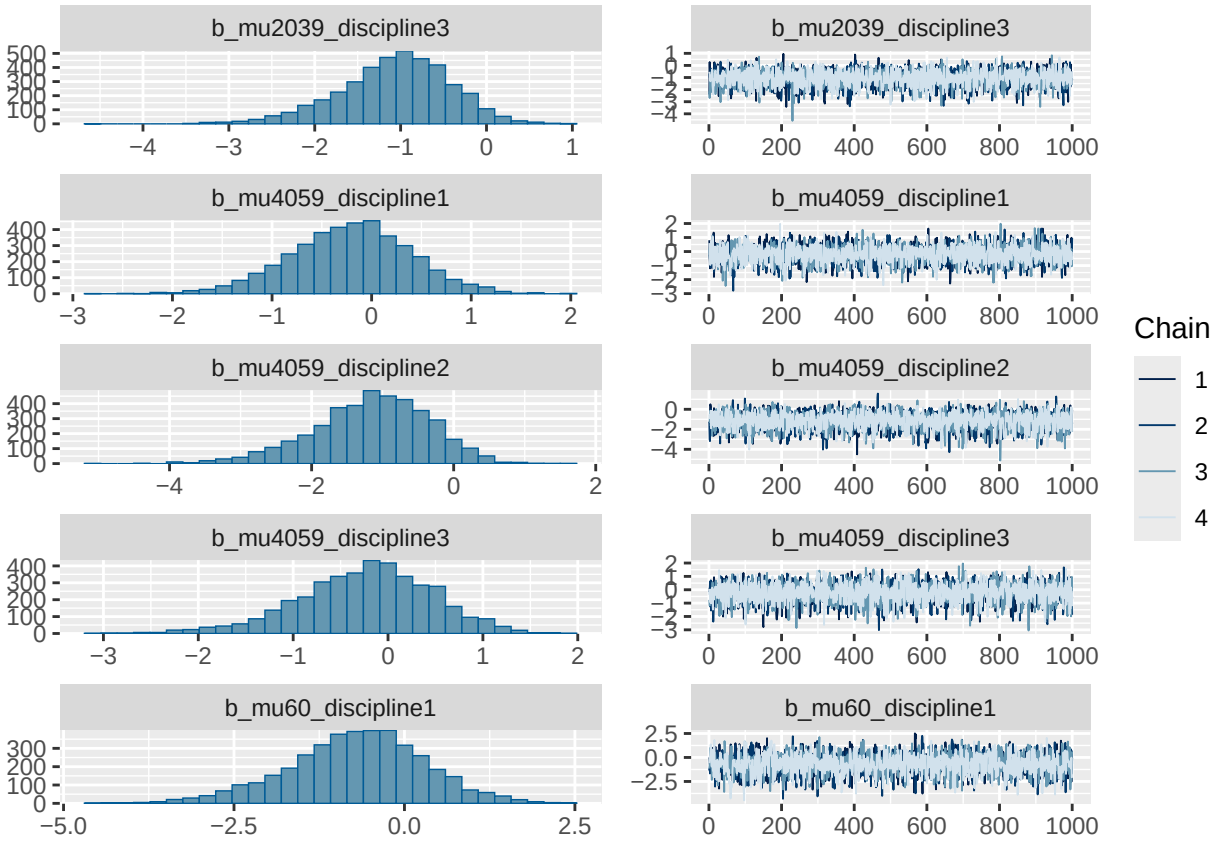
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_sd_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept      -0.74      0.32    -1.40    -0.15 1.00     4097     3170
## mu4059_Intercept      -1.68      0.42    -2.57    -0.92 1.00     3864     3000
## mu60_Intercept        -2.65      0.75    -4.32    -1.42 1.00     2685     2657
## mu2039_discipline1     0.17      0.43    -0.67     1.00 1.00     3780     3085
## mu2039_discipline2    -0.89      0.57    -2.08     0.19 1.00     3747     2692
## mu2039_discipline3    -1.07      0.65    -2.50     0.08 1.00     3868     2766
## mu4059_discipline1    -0.21      0.61    -1.44     0.96 1.00     3756     2340
## mu4059_discipline2    -1.17      0.82    -2.94     0.27 1.00     3462     2865
## mu4059_discipline3    -0.24      0.71    -1.80     1.05 1.00     3469     2549
## mu60_discipline1      -0.70      1.00    -2.81     1.22 1.00     3080     2473
## mu60_discipline2      -0.53      1.00    -2.61     1.27 1.00     3258     2589
## mu60_discipline3      -2.45      1.54    -5.93     0.11 1.00     2767     1974
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

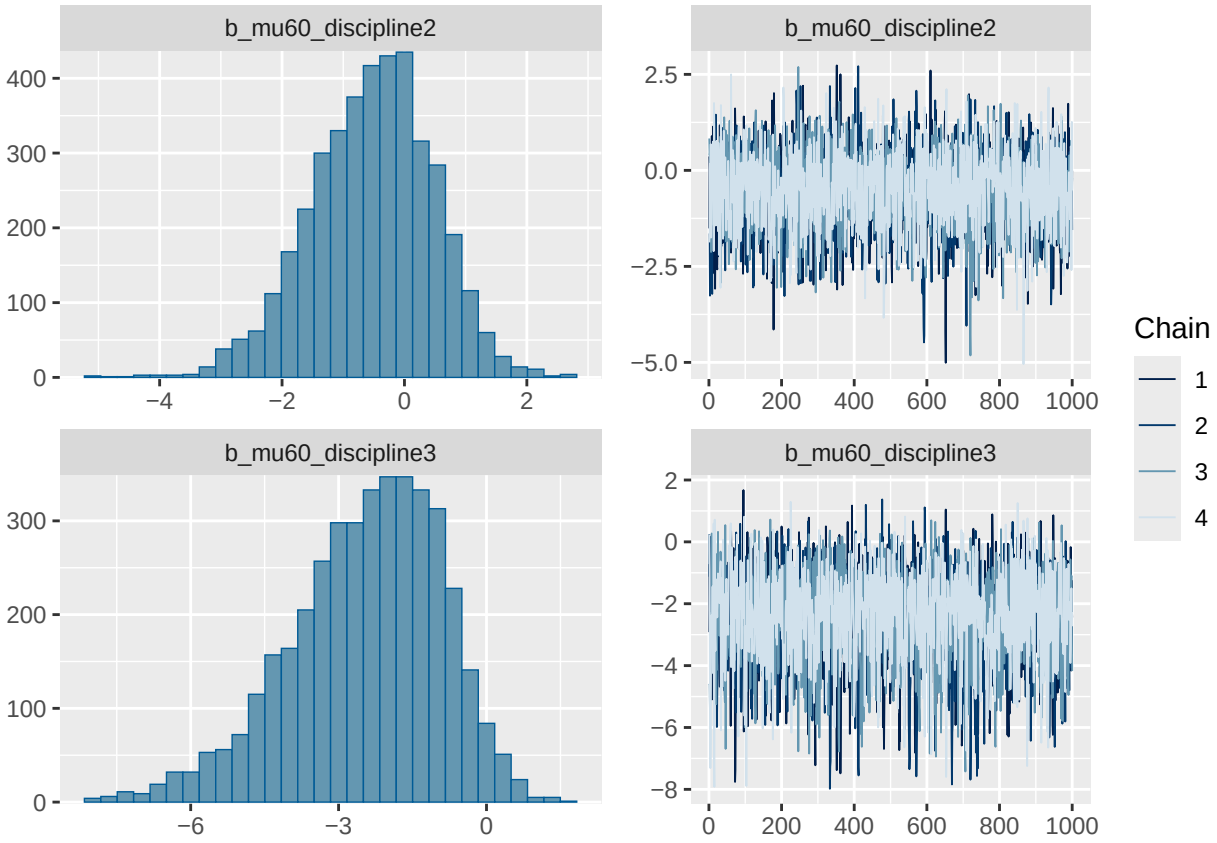
```
#Posterior predictive check:
pp_check(brm.mn.studydesign, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.studydesign)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
Chain
1
2
3
4


b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4



The summary (`summary(brm.mn.studydesign)`) indicates that the model converged well (based on Rhat) and has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.studydesign, type = "bars", ndraws = 100)`) indicates sufficient model fit.

The traces and distributions of estimates on the latent scale (`plot(brm.mn.studydesign)`) also indicate good model behaviour.

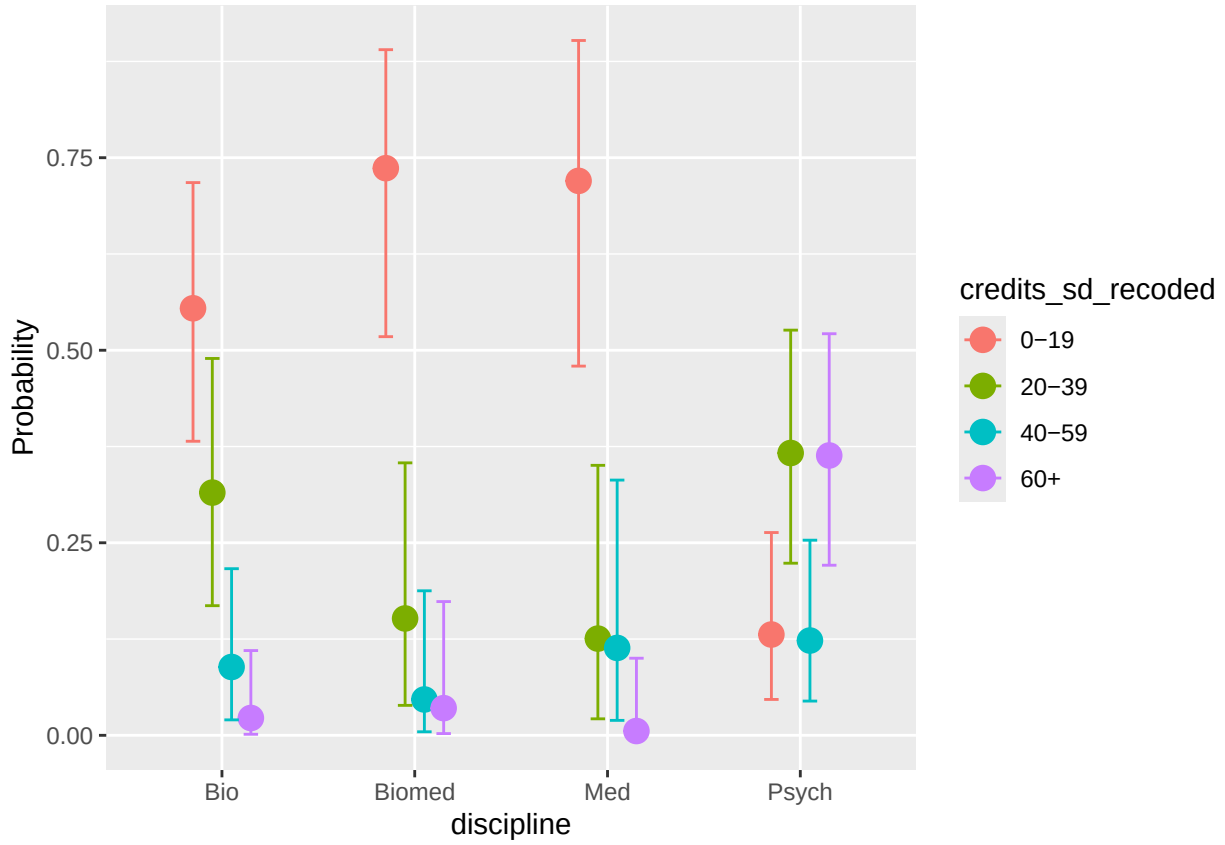
Now we obtain the model estimates:

```
eff_brm.mn.studydesign <- conditional_effects(brm.mn.studydesign, categorical = TRUE)
eff_brm.mn.studydesign[[1]]
```

```
##    discipline credits_sd_recoded cond__ cats__ effect1__ effect2__  estimate__
## 1         Bio               0-19      1   0-19       Bio      0-19 0.554495432
## 2      Biomed               0-19      1   0-19    Biomed      0-19 0.736402417
## 3         Med               0-19      1   0-19       Med      0-19 0.719961557
## 4       Psych               0-19      1   0-19     Psych      0-19 0.130867911
## 5         Bio               0-19      1  20-39       Bio     20-39 0.315161167
## 6      Biomed               0-19      1  20-39    Biomed     20-39 0.151744513
## 7         Med               0-19      1  20-39       Med     20-39 0.125588237
## 8       Psych               0-19      1  20-39     Psych     20-39 0.366572958
## 9         Bio               0-19      1  40-59       Bio     40-59 0.088736441
## 10     Biomed               0-19      1  40-59    Biomed     40-59 0.046531787
## 11        Med               0-19      1  40-59       Med     40-59 0.113468686
## 12      Psych               0-19      1  40-59     Psych     40-59 0.123067020
## 13        Bio               0-19      1    60+       Bio       60+ 0.022494028
```

```
## 14     Biomed              0-19      1    60+    Biomed       60+ 0.035292263
## 15        Med              0-19      1    60+       Med       60+ 0.005562305
## 16      Psych              0-19      1    60+     Psych       60+ 0.363350150
##             se__       lower__   upper__
## 1   0.090980732 3.818124e-01 0.7177157
## 2   0.098909033 5.176463e-01 0.8902722
## 3   0.117676015 4.794207e-01 0.9022342
## 4   0.055783222 4.657382e-02 0.2633146
## 5   0.084856309 1.683175e-01 0.4893278
## 6   0.080755149 3.888741e-02 0.3537889
## 7   0.080892481 2.141974e-02 0.3505775
## 8   0.079633737 2.234849e-01 0.5261190
## 9   0.049184793 2.004293e-02 0.2163595
## 10  0.039581027 4.576272e-03 0.1876306
## 11  0.078631954 1.938321e-02 0.3314007
## 12  0.054800102 4.445547e-02 0.2533645
## 13  0.022115081 1.341990e-03 0.1100640
## 14  0.034658539 2.264731e-03 0.1736719
## 15  0.007635368 3.525637e-05 0.1001157
## 16  0.078763508 2.208442e-01 0.5214428
```

```
print(eff_brm.mn.studydesign)
```



Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)
```

```
brm.mn_fixed <- tidy(brm.mn.studydesign, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %>%
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.studydesign, effects = "fixed", conf.level =
## 0.95, : some parameter names contain underscores: term naming may be
## unreliable!
```

```
brm.mn_fixed
```

```
## # A tibble: 12 x 8
##    effect component term          estimate std.error conf.low conf.high ci_width
##    <chr>  <chr>     <chr>            <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
##  1 fixed  cond      mu2039_Inter~   -0.739     0.317   -1.40    -0.155      1.25
##  2 fixed  cond      mu4059_Inter~   -1.68      0.418   -2.57    -0.920      1.65
##  3 fixed  cond      mu60_Interce~   -2.65      0.748   -4.32    -1.42       2.90
##  4 fixed  cond      mu2039_disci~    0.166     0.427   -0.671    0.995      1.67
##  5 fixed  cond      mu2039_disci~   -0.886     0.565   -2.08     0.190      2.27
##  6 fixed  cond      mu2039_disci~   -1.07      0.651   -2.50     0.0813     2.58
##  7 fixed  cond      mu4059_disci~   -0.209     0.607   -1.44     0.962      2.40
##  8 fixed  cond      mu4059_disci~   -1.17      0.819   -2.94     0.268      3.21
##  9 fixed  cond      mu4059_disci~   -0.240     0.706   -1.80     1.05       2.84
## 10 fixed  cond      mu60_discipl~   -0.705     1.00    -2.81     1.22       4.03
## 11 fixed  cond      mu60_discipl~   -0.529     1.00    -2.61     1.27       3.88
## 12 fixed  cond      mu60_discipl~   -2.45      1.54    -5.93     0.108      6.04
```

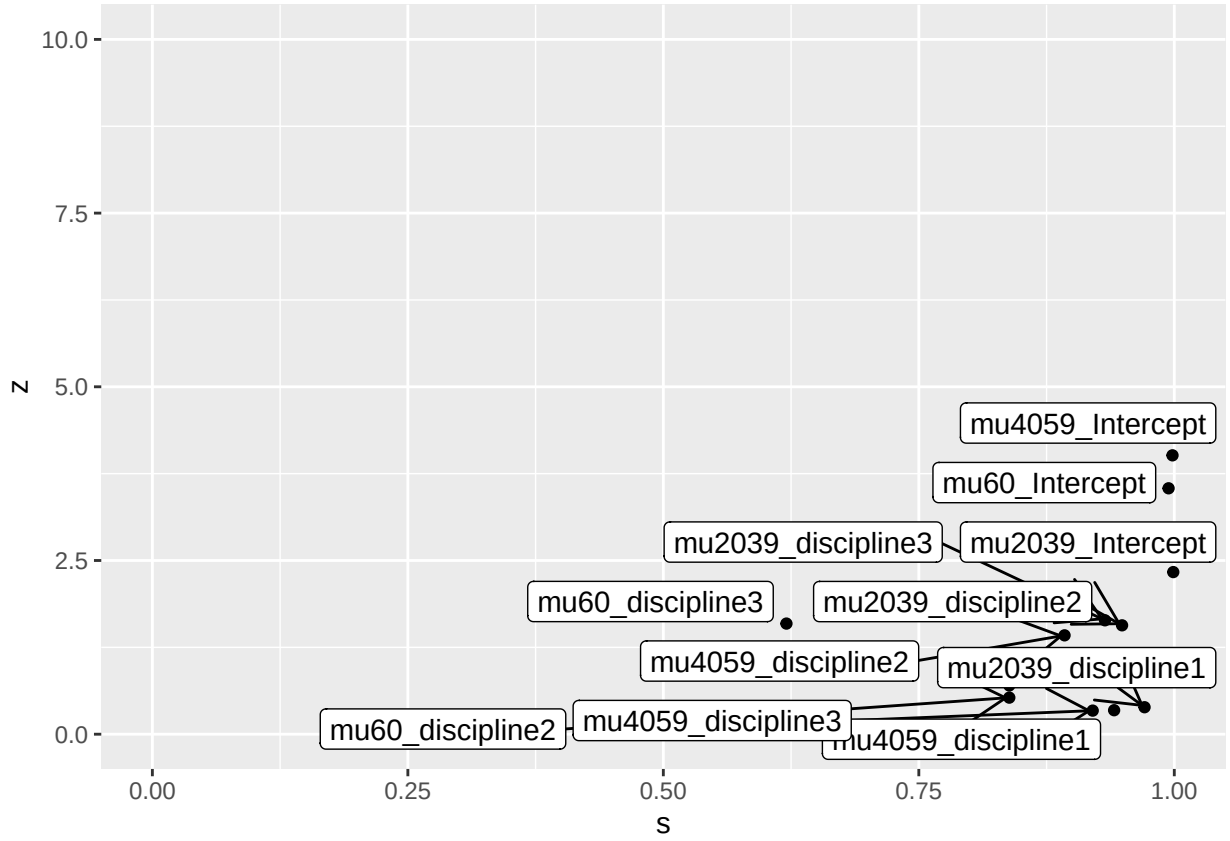
```
brm.mn_fixed_reduced <- brm.mn_fixed %>%
  select(
    effect, component, term, estimate
  )
brm.mn_fixed_reduced
```

```
## # A tibble: 12 x 4
##    effect component term                estimate
##    <chr>  <chr>     <chr>                  <dbl>
##  1 fixed  cond      mu2039_Intercept      -0.739
##  2 fixed  cond      mu4059_Intercept      -1.68
##  3 fixed  cond      mu60_Intercept        -2.65
##  4 fixed  cond      mu2039_discipline1     0.166
##  5 fixed  cond      mu2039_discipline2    -0.886
##  6 fixed  cond      mu2039_discipline3    -1.07
##  7 fixed  cond      mu4059_discipline1    -0.209
##  8 fixed  cond      mu4059_discipline2    -1.17
##  9 fixed  cond      mu4059_discipline3    -0.240
## 10 fixed  cond      mu60_discipline1      -0.705
## 11 fixed  cond      mu60_discipline2      -0.529
## 12 fixed  cond      mu60_discipline3      -2.45
```

```
#We have 12 fixed effects; the first 3 are intercepts, and the last 9 are coefficients (b)
#Also, the 3 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 9 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn_fixed %>%
```

```
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  ) %>%
  ggplot(aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

```
## Warning: ggrepel: 1 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Credits: Data Analysis

Models with 4 categories

recode categories

```
#recode categories
survey_data <-
  survey_data %>%
  mutate(
    credits_da_recoded = fct_collapse(credits_da,
```

```
                                    "0-19" = c("0", "1-9", "10-19"),
                                    "20-39"= c("20-29","30-39"),
                                    "40-59" = c("40-49", "50-59"),
                                    "60+" = c("60-69", "70-79", "80+")))

#structure: ordered factor with 4 levels
glimpse(survey_data$credits_da_recoded)
```

```
##  Ord.factor w/ 4 levels "0-19"<"20-39"<..: 4 3 1 1 2 3 2 1 1 1 ...
```

```
#table: data analysis credits (4 categories) by discipline
table(survey_data$discipline, survey_data$credits_da_recoded)
```

```
##
##          0-19 20-39 40-59 60+
##   Bio      15    10     4   2
##   Biomed   12     5     0   2
##   Med      10     4     1   0
##   Psych     2    12     9  15
```

Multinomial Analysis of Data Analysis Credits

We now fit a multinomial model, using a prior inspired by Gelman et al. (2008) (we use the recommended values for standard deviation, but use normal distributions rather than cauchy).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```
#We obtain the default prior, and then update the prior, below.
get_prior(credits_da_recoded ~ discipline, family = categorical(link = "logit"),
          data = survey_data)
```

```
##                 prior     class        coef group resp   dpar nlpar lb ub
##                (flat)         b                         mu2039
##                (flat)         b discipline1           mu2039
##                (flat)         b discipline2           mu2039
##                (flat)         b discipline3           mu2039
##  student_t(3, 0, 2.5) Intercept                         mu2039
##                (flat)         b                         mu4059
##                (flat)         b discipline1           mu4059
##                (flat)         b discipline2           mu4059
##                (flat)         b discipline3           mu4059
##  student_t(3, 0, 2.5) Intercept                         mu4059
##                (flat)         b                           mu60
##                (flat)         b discipline1               mu60
##                (flat)         b discipline2               mu60
##                (flat)         b discipline3               mu60
##  student_t(3, 0, 2.5) Intercept                           mu60
##       source
##      default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##      default
##      default
##  (vectorized)
```

```
##  (vectorized)
##  (vectorized)
##       default
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
```

```
#Our prior, following Gelman et al. (2008), but we use a normal (not cauchy) distribution:
priors.dataanalysis <- c(
  prior(normal(0, 10), class = "Intercept", dpar = "mu2039"),
  prior(normal(0, 10), class = "Intercept", dpar = "mu4059"),
  prior(normal(0, 10), class = "Intercept", dpar = "mu60"),
  prior(normal(0, 2.5), class = "b", dpar = "mu2039"),
  prior(normal(0, 2.5), class = "b", dpar = "mu4059"),
  prior(normal(0, 2.5), class = "b", dpar = "mu60")
)


contrasts(survey_data$discipline) =contr.sum(4)
contrasts(survey_data$discipline)
```

```
##         [,1] [,2] [,3]
## Bio        1    0    0
## Biomed     0    1    0
## Med        0    0    1
## Psych     -1   -1   -1
```

```
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1

#Prior predictive check:
brm.formula1.da <- brmsformula(credits_da_recoded ~ discipline,
                               family = categorical(link = "logit"))

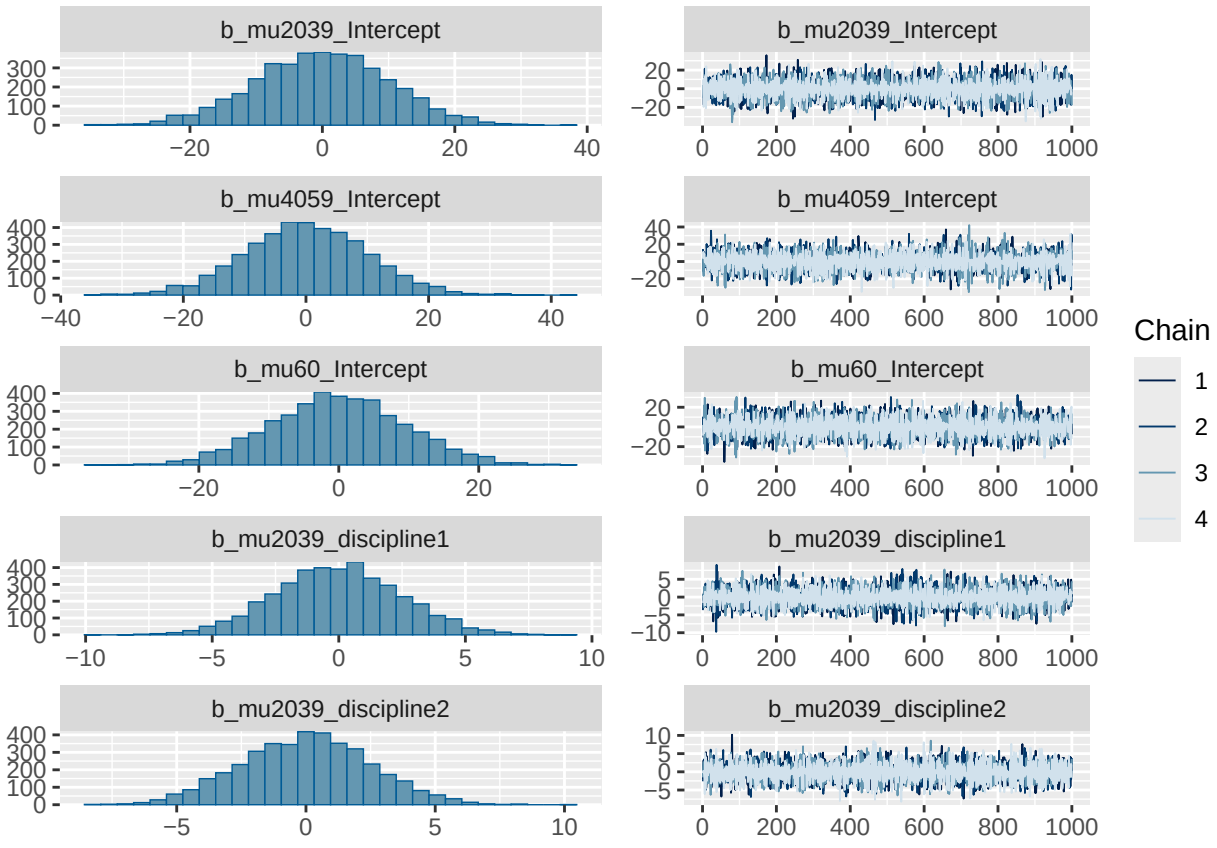
brm.mn.dataanalysis.priorpc <- brm(
  brm.formula1.da,
  prior = priors.dataanalysis,
  sample_prior = "only",
  data = survey_data,
  file = here("store/lists", "brm.mn.dataanalysis.priorpc"))

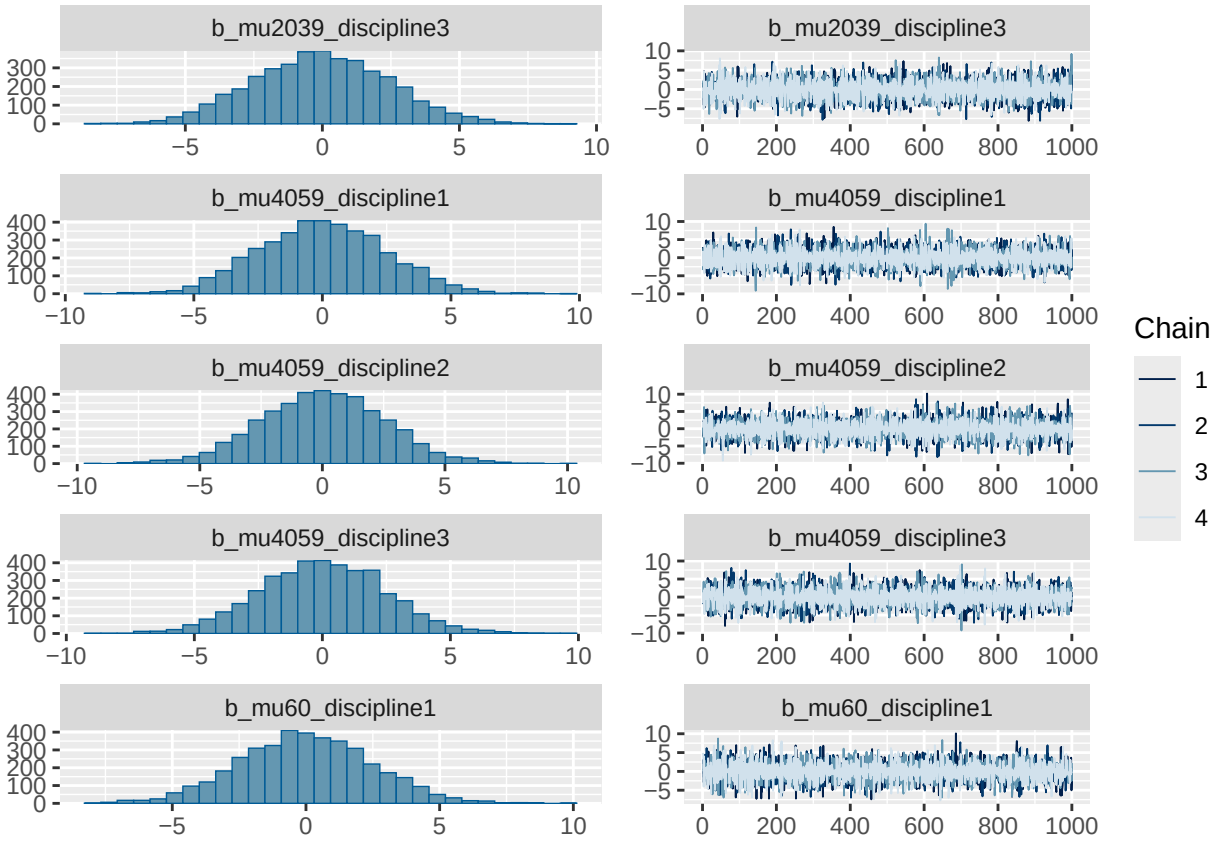
summary(brm.mn.dataanalysis.priorpc)
```

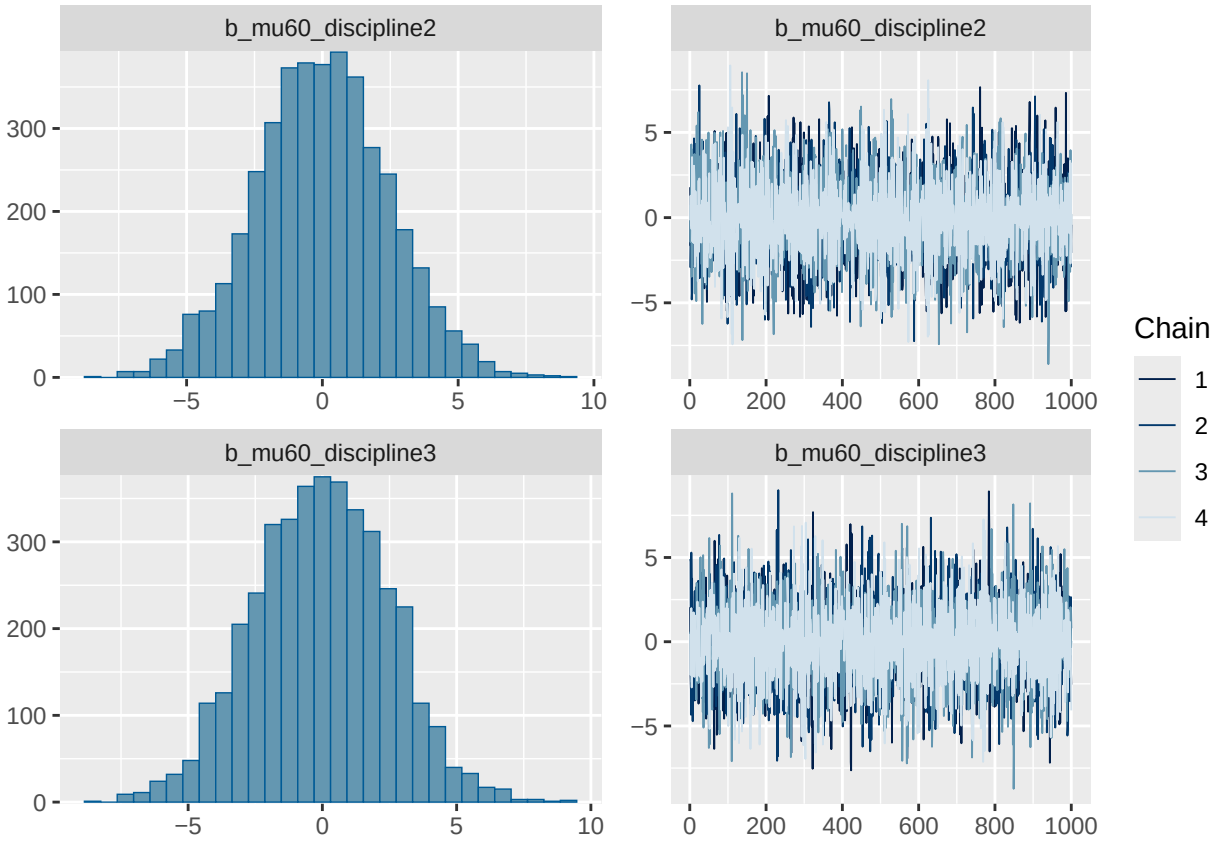
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_da_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
```

```
## Regression Coefficients:
##                     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept       -0.03     10.14   -20.70    20.03 1.00     9214     2974
## mu4059_Intercept        0.04     10.07   -20.25    19.99 1.00     9304     3197
## mu60_Intercept         -0.03      9.59   -18.58    18.95 1.00     8157     3116
## mu2039_discipline1     -0.03      2.49    -4.88     4.81 1.00     7528     3281
## mu2039_discipline2      0.00      2.50    -4.83     4.90 1.00     9204     2533
## mu2039_discipline3     -0.04      2.49    -4.78     4.79 1.00     9662     3080
## mu4059_discipline1      0.02      2.47    -4.65     4.81 1.00     9428     3281
## mu4059_discipline2      0.03      2.53    -4.97     5.01 1.00     8637     2999
## mu4059_discipline3      0.01      2.45    -4.79     4.82 1.00     9643     2988
## mu60_discipline1       -0.02      2.48    -4.87     4.73 1.00     8594     3096
## mu60_discipline2       -0.00      2.44    -4.85     4.80 1.00     7523     2819
## mu60_discipline3       -0.03      2.51    -4.88     4.76 1.00     8074     2738
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

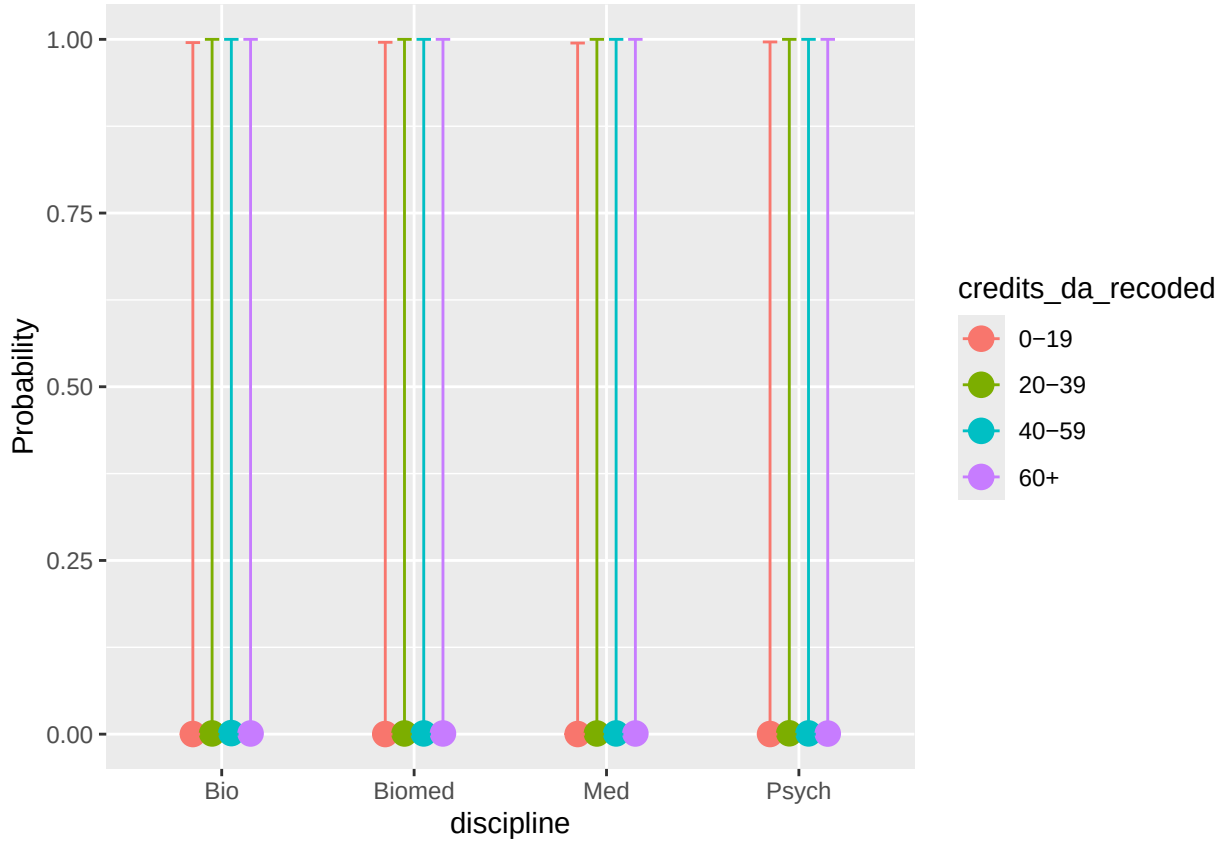
```
plot(brm.mn.dataanalysis.priorpc)
```




b_mu2039_discipline3
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline2
b_mu4059_discipline3
b_mu4059_discipline3
b_mu60_discipline1
b_mu60_discipline1
Chain
1
2
3
4




```
#Prior predictive check outout:
conditional_effects(brm.mn.dataanalysis.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.dataanalysis.priorpc)` and `plot(brm.mn.dataanalysis.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman-ish prior:
brm.mn.dataanalysis <- brm(credits_da_recoded ~ discipline,
                           family = categorical(link = "logit"),
                           prior = priors.dataanalysis,
                           data = survey_data,
                           file = here("store/models", "brm.mn.dataanalysis"))

#Examine prior that we specified:
prior_summary(brm.mn.dataanalysis)
```

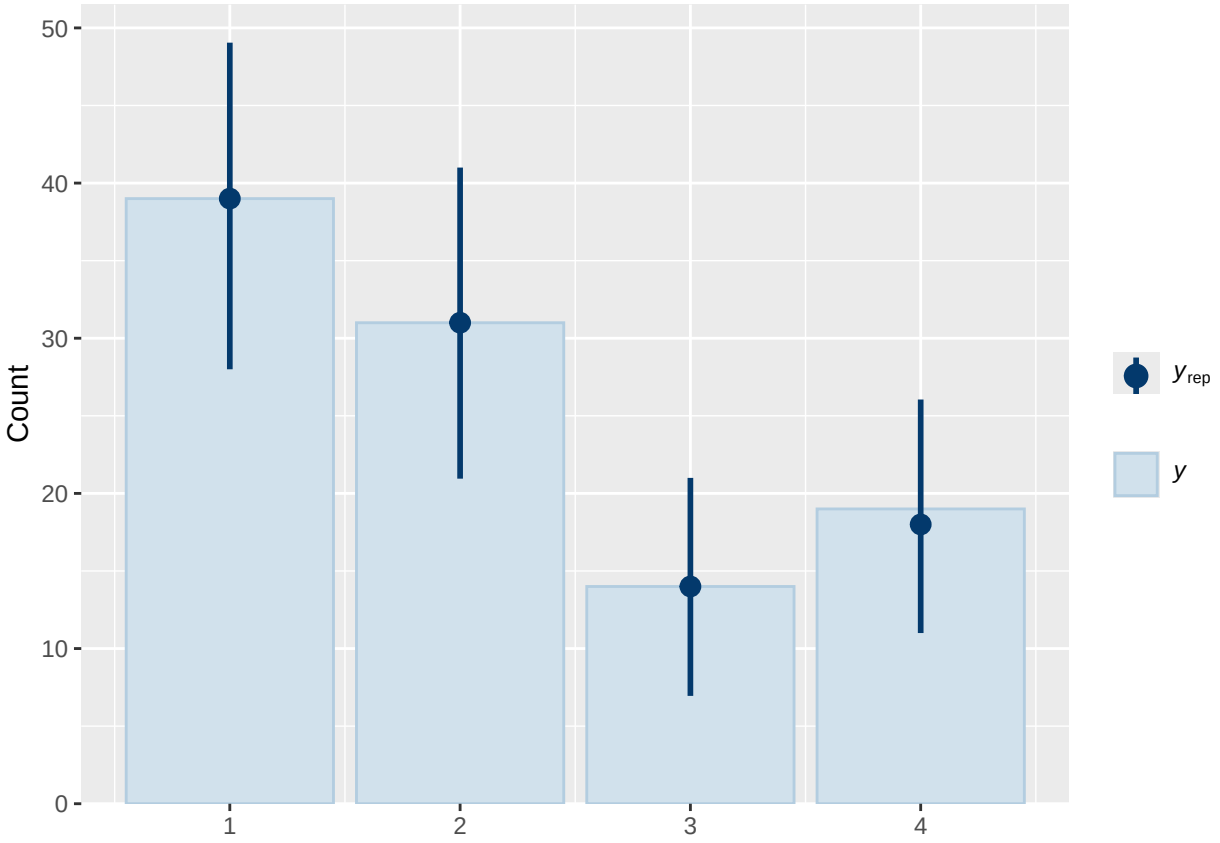
```
##           prior     class        coef group resp   dpar nlpar lb ub
##  normal(0, 2.5)         b                       mu2039
##  normal(0, 2.5)         b disciplineBio         mu2039
##  normal(0, 2.5)         b disciplineBiomed      mu2039
##  normal(0, 2.5)         b disciplineMed         mu2039
##  normal(0, 2.5)         b disciplinePsych       mu2039
##  normal(0, 2.5)         b                       mu4059
##  normal(0, 2.5)         b disciplineBio         mu4059
##  normal(0, 2.5)         b disciplineBiomed      mu4059
##  normal(0, 2.5)         b disciplineMed         mu4059
##  normal(0, 2.5)         b disciplinePsych       mu4059
##  normal(0, 2.5)         b                         mu60
##  normal(0, 2.5)         b disciplineBio           mu60
##  normal(0, 2.5)         b disciplineBiomed        mu60
```

```
##  normal(0, 2.5)         b discipline3              mu60
##   normal(0, 10) Intercept                         mu2039
##   normal(0, 10) Intercept                         mu4059
##   normal(0, 10) Intercept                           mu60
##       source
##         user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##         user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##         user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##         user
##         user
##         user
```

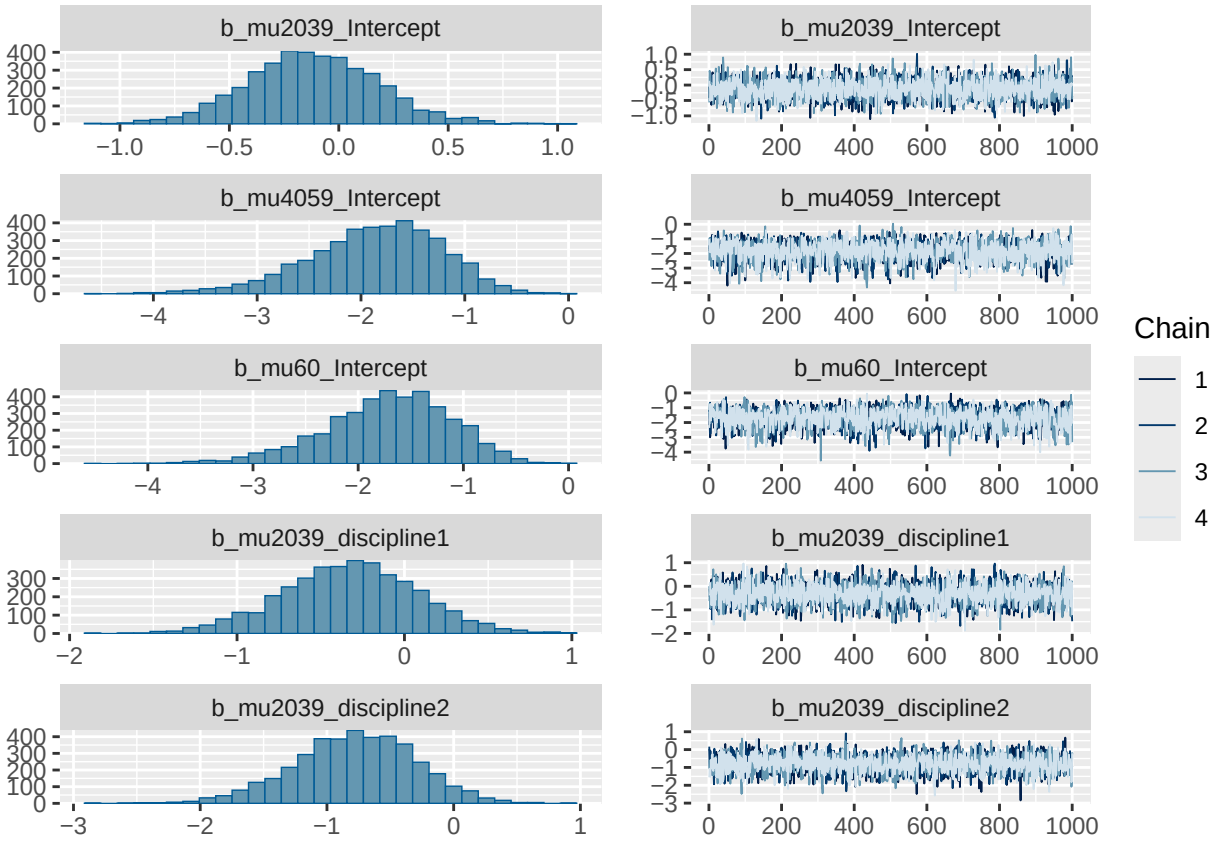
```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.dataanalysis)
```

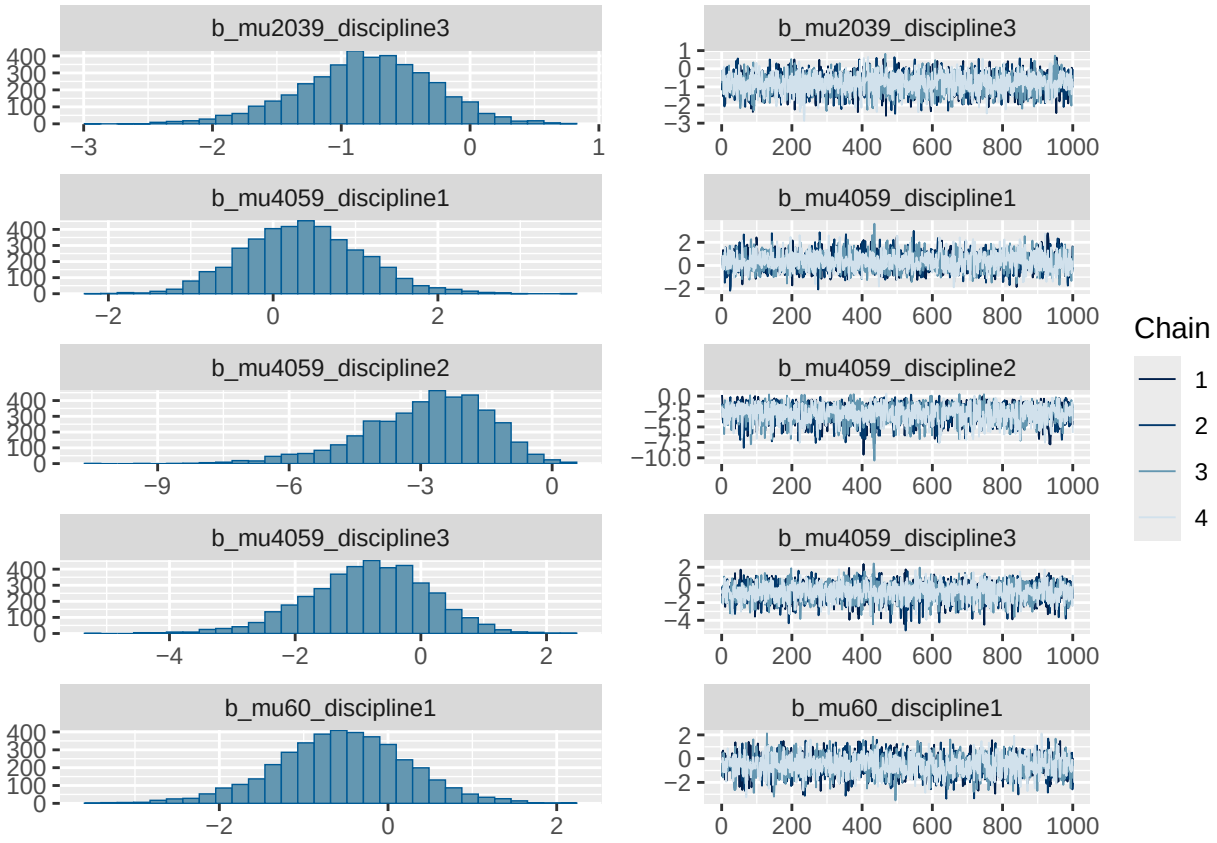
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_da_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept      -0.12      0.30    -0.70     0.47 1.00     3134     2808
## mu4059_Intercept      -1.86      0.64    -3.28    -0.77 1.00     2209     2148
## mu60_Intercept        -1.72      0.60    -3.04    -0.67 1.00     2257     2158
## mu2039_discipline1    -0.33      0.41    -1.15     0.47 1.00     2823     2706
## mu2039_discipline2    -0.78      0.48    -1.76     0.12 1.00     3446     2422
## mu2039_discipline3    -0.81      0.52    -1.90     0.14 1.00     3339     2790
## mu4059_discipline1     0.37      0.73    -0.99     1.87 1.00     2483     2526
## mu4059_discipline2    -2.87      1.43    -6.20    -0.60 1.00     2298     1736
## mu4059_discipline3    -0.85      0.97    -2.90     0.92 1.00     2810     2486
## mu60_discipline1      -0.55      0.79    -2.18     0.97 1.00     2219     2470
## mu60_discipline2      -0.33      0.80    -1.91     1.20 1.00     2705     2699
## mu60_discipline3      -2.86      1.39    -6.03    -0.65 1.00     2230     1871
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

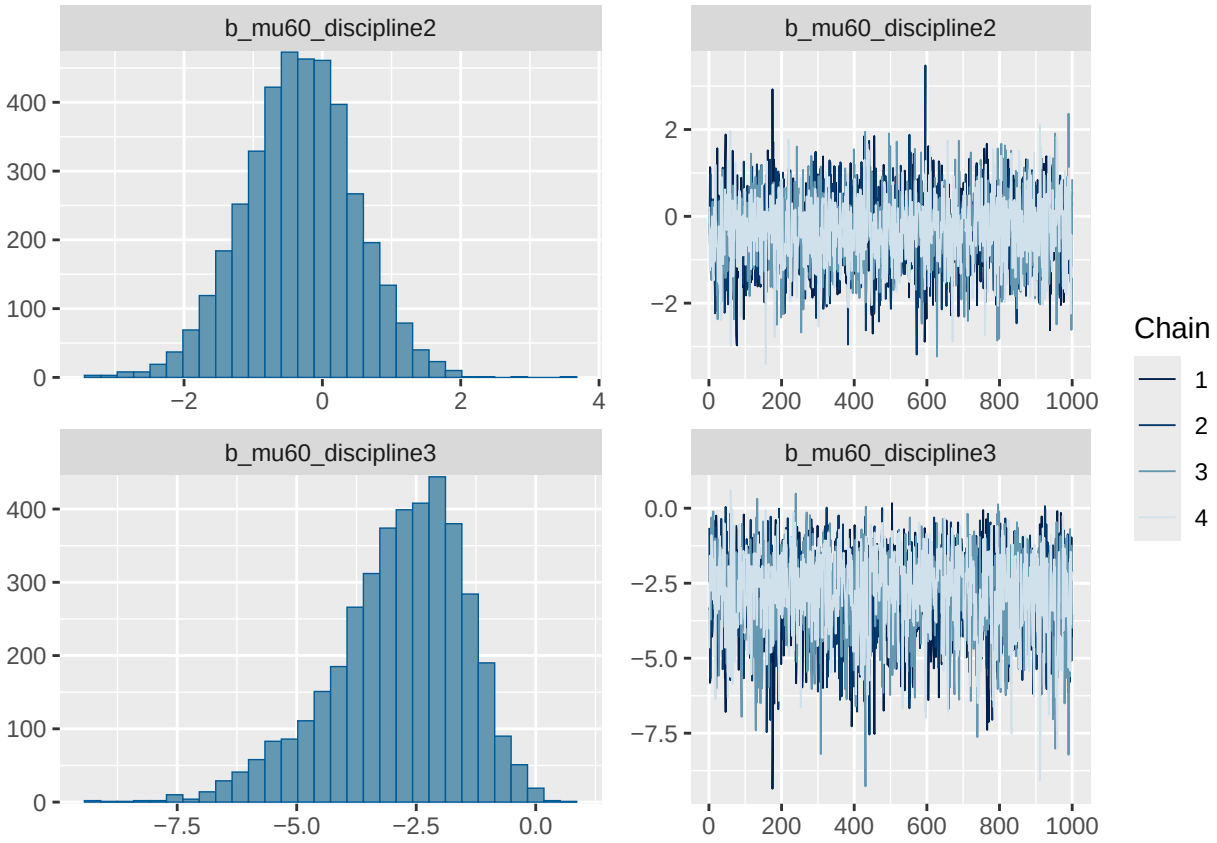
```
#Posterior predictive check:
pp_check(brm.mn.dataanalysis, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.dataanalysis)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
Chain
1
2
3
4


b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4



The summary (`summary(brm.mn.dataanalysis)`) indicates that the model converged well (based on Rhat) and has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.dataanalysis, type = "bars", ndraws = 100)`) indicates sufficient model fit.

The traces and distributions of estimates on the latent scale (`plot(brm.mn.dataanalysis)`) also indicate good model behaviour.

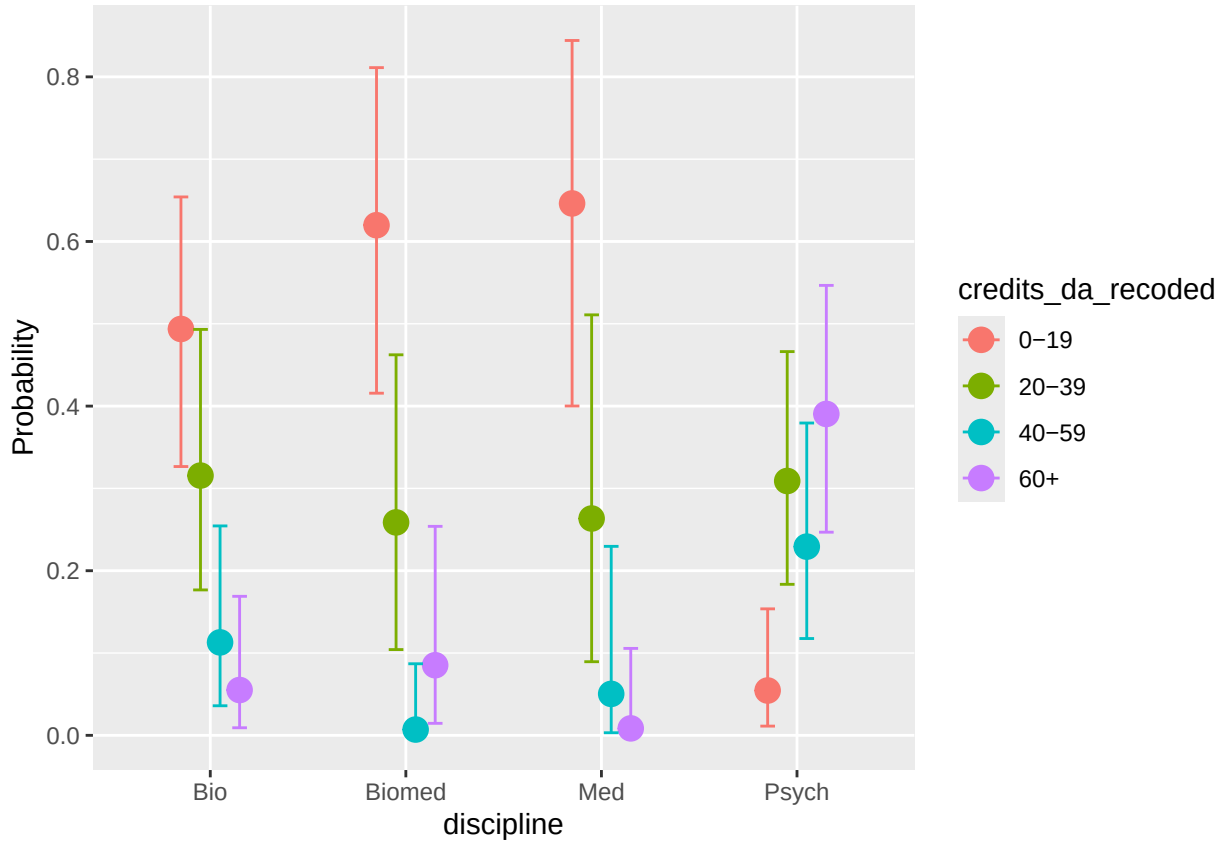
Now we obtain the model estimates:

```
eff_brm.mn.dataanalysis <- conditional_effects(brm.mn.dataanalysis, categorical = TRUE)
eff_brm.mn.dataanalysis[[1]]
```

```
##    discipline credits_da_recoded cond__ cats__ effect1__ effect2__  estimate__
## 1         Bio               0-19      1   0-19       Bio      0-19 0.493711489
## 2      Biomed               0-19      1   0-19    Biomed      0-19 0.619782552
## 3         Med               0-19      1   0-19       Med      0-19 0.646203663
## 4       Psych               0-19      1   0-19     Psych      0-19 0.054446312
## 5         Bio               0-19      1  20-39       Bio     20-39 0.315587443
## 6      Biomed               0-19      1  20-39    Biomed     20-39 0.258694734
## 7         Med               0-19      1  20-39       Med     20-39 0.263437738
## 8       Psych               0-19      1  20-39     Psych     20-39 0.309021454
## 9         Bio               0-19      1  40-59       Bio     40-59 0.112930194
## 10     Biomed               0-19      1  40-59    Biomed     40-59 0.007028225
## 11        Med               0-19      1  40-59       Med     40-59 0.050358928
## 12      Psych               0-19      1  40-59     Psych     40-59 0.229262108
## 13        Bio               0-19      1    60+       Bio       60+ 0.055061795
```

```
## 14     Biomed              0-19      1    60+    Biomed       60+ 0.085198453
## 15        Med              0-19      1    60+       Med       60+ 0.008448973
## 16      Psych              0-19      1    60+     Psych       60+ 0.390477106
##             se__      lower__   upper__
## 1   0.085840856 3.265750e-01 0.6541312
## 2   0.105466809 4.156760e-01 0.8112096
## 3   0.117224458 4.001186e-01 0.8442054
## 4   0.034217725 1.121355e-02 0.1536669
## 5   0.081348014 1.766543e-01 0.4931739
## 6   0.099800484 1.041784e-01 0.4623235
## 7   0.112113394 8.945881e-02 0.5108644
## 8   0.073927968 1.834578e-01 0.4662058
## 9   0.054603434 3.591748e-02 0.2544229
## 10 0.009358613 6.095001e-05 0.0869425
## 11 0.047494124 3.130164e-03 0.2296010
## 12 0.063803233 1.176194e-01 0.3794330
## 13 0.038451316 9.194758e-03 0.1688862
## 14 0.059357996 1.457633e-02 0.2539129
## 15 0.011078862 8.410045e-05 0.1056258
## 16 0.077732304 2.468189e-01 0.5466544
```

```
print(eff_brm.mn.dataanalysis)
```



Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)
```

```
brm.mn_fixed <- tidy(brm.mn.dataanalysis, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.dataanalysis, effects = "fixed", conf.level =
## 0.95, : some parameter names contain underscores: term naming may be
## unreliable!
```

```
brm.mn_fixed
```

```
## # A tibble: 12 x 8
##    effect component term          estimate std.error conf.low conf.high ci_width
##    <chr>  <chr>     <chr>            <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
##  1 fixed  cond      mu2039_Inter~   -0.120     0.298   -0.701     0.471     1.17
##  2 fixed  cond      mu4059_Inter~   -1.86      0.644   -3.28     -0.767     2.52
##  3 fixed  cond      mu60_Interce~   -1.72      0.605   -3.04     -0.668     2.37
##  4 fixed  cond      mu2039_disci~   -0.327     0.407   -1.15      0.466     1.61
##  5 fixed  cond      mu2039_disci~   -0.780     0.478   -1.76      0.124     1.89
##  6 fixed  cond      mu2039_disci~   -0.807     0.516   -1.90      0.144     2.04
##  7 fixed  cond      mu4059_disci~    0.366     0.727   -0.985     1.87      2.86
##  8 fixed  cond      mu4059_disci~   -2.87      1.43    -6.20     -0.597     5.61
##  9 fixed  cond      mu4059_disci~   -0.847     0.966   -2.90      0.918     3.81
## 10 fixed  cond      mu60_discipl~   -0.552     0.791   -2.18      0.975     3.15
## 11 fixed  cond      mu60_discipl~   -0.329     0.802   -1.91      1.20      3.11
## 12 fixed  cond      mu60_discipl~   -2.86      1.39    -6.03     -0.650     5.38
```

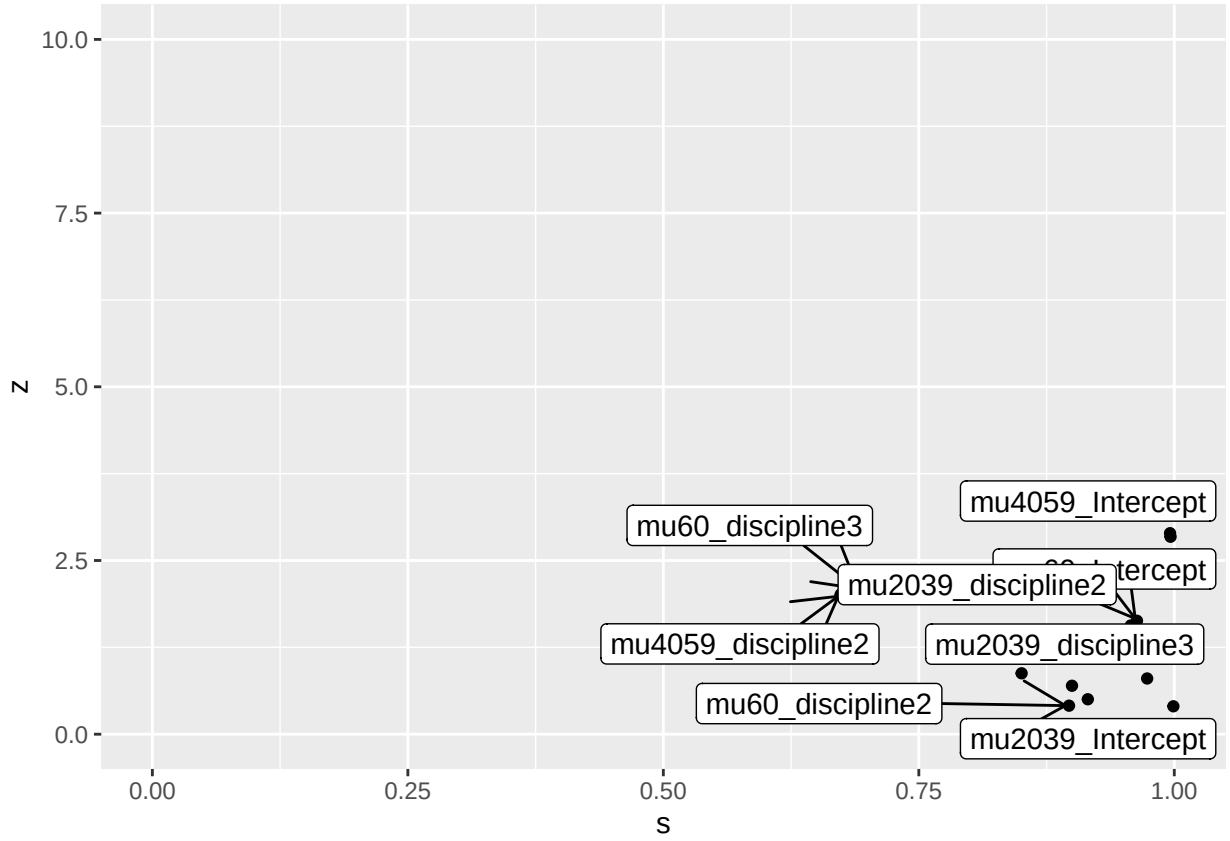
```
brm.mn_fixed_reduced <- brm.mn_fixed %>%
  select(
    effect, component, term, estimate
  )
brm.mn_fixed_reduced
```

```
## # A tibble: 12 x 4
##    effect component term               estimate
##    <chr>  <chr>     <chr>                 <dbl>
##  1 fixed  cond      mu2039_Intercept     -0.120
##  2 fixed  cond      mu4059_Intercept     -1.86
##  3 fixed  cond      mu60_Intercept       -1.72
##  4 fixed  cond      mu2039_discipline1   -0.327
##  5 fixed  cond      mu2039_discipline2   -0.780
##  6 fixed  cond      mu2039_discipline3   -0.807
##  7 fixed  cond      mu4059_discipline1    0.366
##  8 fixed  cond      mu4059_discipline2   -2.87
##  9 fixed  cond      mu4059_discipline3   -0.847
## 10 fixed  cond      mu60_discipline1     -0.552
## 11 fixed  cond      mu60_discipline2     -0.329
## 12 fixed  cond      mu60_discipline3     -2.86
```

```
#We have 12 fixed effects; the first 3 are intercepts, and the last 9 are coefficients (b)
#Also, the 3 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 9 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn_fixed %>%
```

```
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  ) %>%
  ggplot(aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

```
## Warning: ggrepel: 4 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Plots

```
plot_colour <- "skyblue4"

plot_theme <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.x = element_blank(),
    axis.text.x = element_text(size= 10),
    axis.title.y = element_text(size= 14),
    axis.text.y = element_markdown(size = 12, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 14),
```

```
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")

plot_theme_2 <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.y = element_blank(),
    axis.text.x = element_text(size= 10),
    axis.title.x = element_text(size= 14),
    axis.text.y = element_markdown(size = 12, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 14),
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")
```

```
study_design_estimates <- eff_brm.mn.studydesign[[1]] %>%
  clean_names() %>%
  rename(
    credits_sd = cats,
    probability_estimate = estimate) %>%
  select(discipline, credits_sd, probability_estimate, lower, upper)

study_design_estimates
```

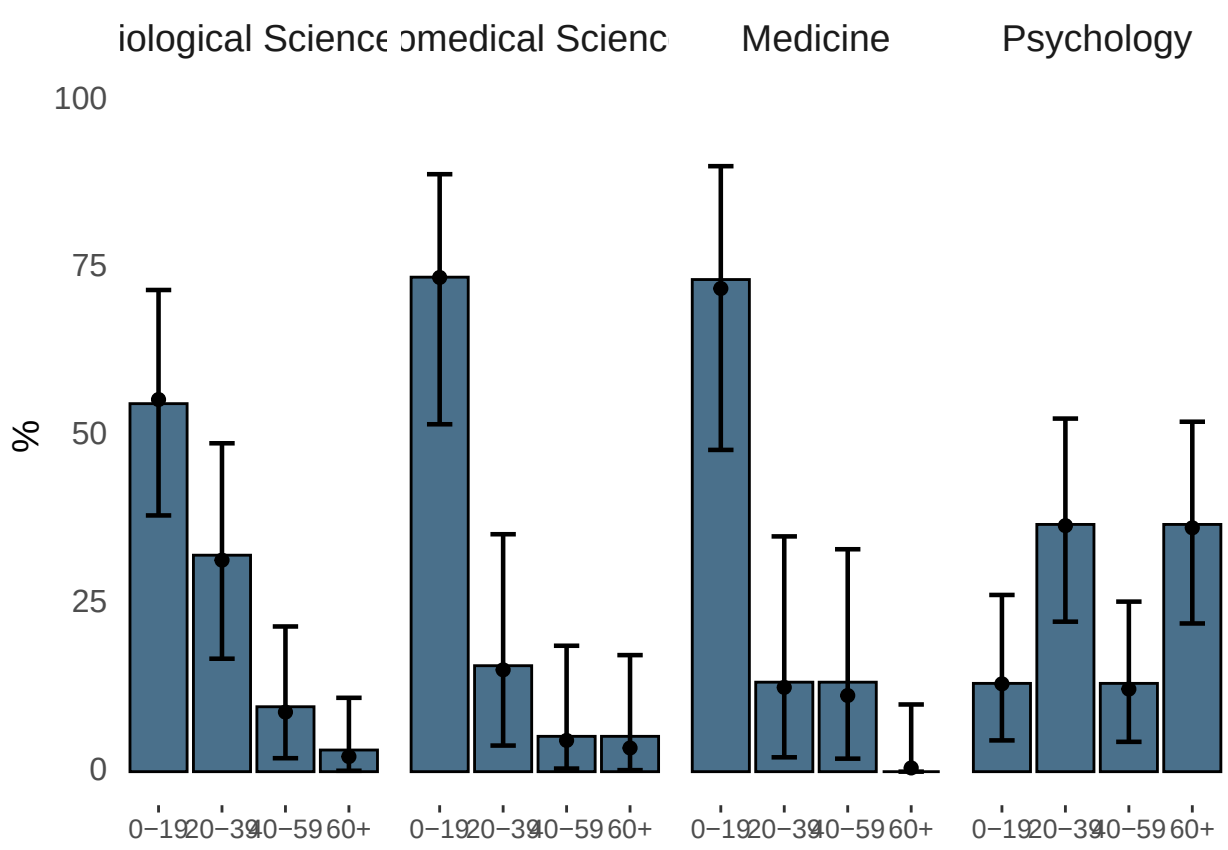
```
##    discipline credits_sd probability_estimate        lower     upper
## 1         Bio       0-19          0.554495432 3.818124e-01 0.7177157
## 2      Biomed       0-19          0.736402417 5.176463e-01 0.8902722
## 3         Med       0-19          0.719961557 4.794207e-01 0.9022342
## 4       Psych       0-19          0.130867911 4.657382e-02 0.2633146
## 5         Bio      20-39          0.315161167 1.683175e-01 0.4893278
## 6      Biomed      20-39          0.151744513 3.888741e-02 0.3537889
## 7         Med      20-39          0.125588237 2.141974e-02 0.3505775
## 8       Psych      20-39          0.366572958 2.234849e-01 0.5261190
## 9         Bio      40-59          0.088736441 2.004293e-02 0.2163595
## 10     Biomed      40-59          0.046531787 4.576272e-03 0.1876306
## 11        Med      40-59          0.113468686 1.938321e-02 0.3314007
## 12      Psych      40-59          0.123067020 4.445547e-02 0.2533645
## 13        Bio        60+          0.022494028 1.341990e-03 0.1100640
## 14     Biomed        60+          0.035292263 2.264731e-03 0.1736719
## 15        Med        60+          0.005562305 3.525637e-05 0.1001157
## 16      Psych        60+          0.363350150 2.208442e-01 0.5214428
```

```
survey_table <- table(survey_data$credits_sd_recoded, survey_data$discipline)
survey_table <- prop.table(survey_table, margin = 2)
survey_table <- as.data.frame(survey_table)
survey_ggplot <- survey_table %>%
  rename(
    credits_sd = Var1,
```

```
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
study_design_plot <- ggplot() +
  geom_col(
    data = survey_ggplot,
    aes(x = credits_sd, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = study_design_estimates,
    aes(x = credits_sd, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = study_design_estimates,
    aes(x = credits_sd, y = 100*probability_estimate),
    size = 2) +
  # #coord_flip() +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  plot_theme


study_design_plot
```

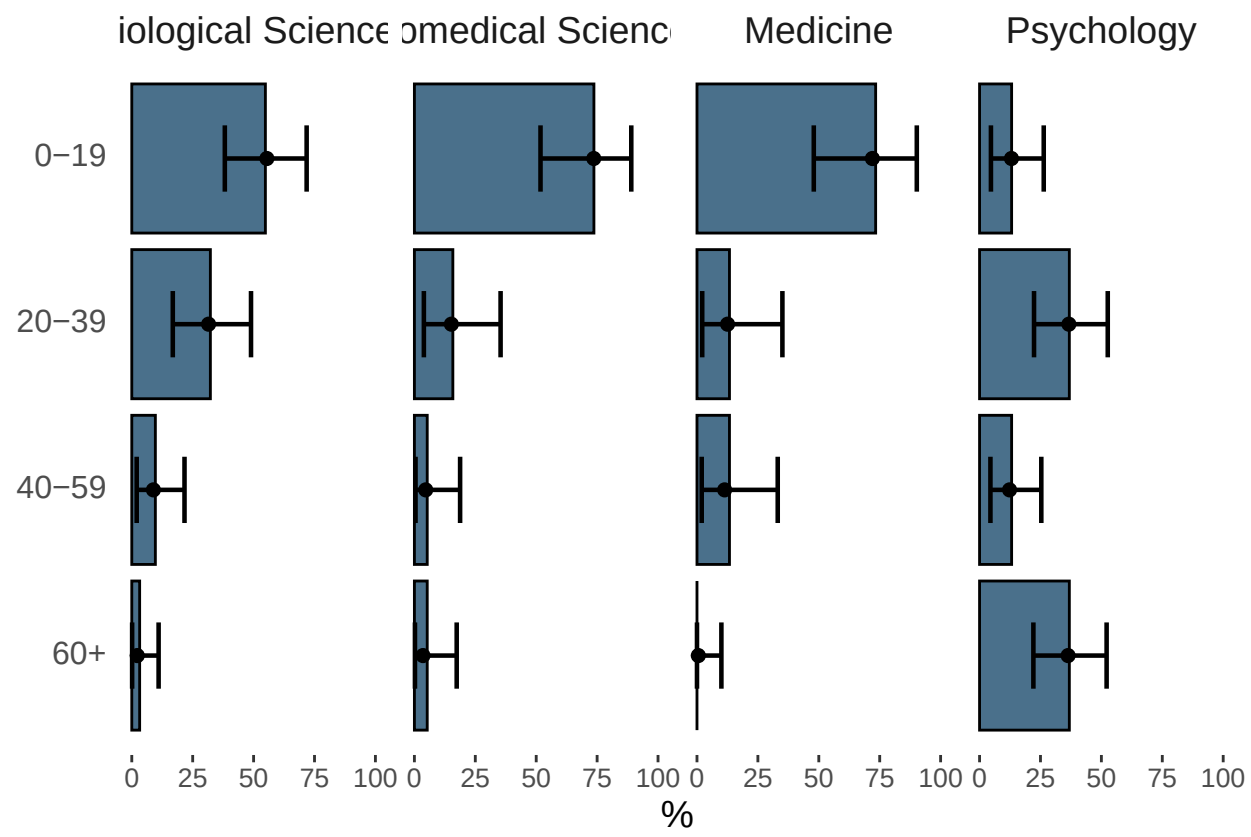



```
ggsave(
  plot = study_design_plot,
  here("figures", "credits_sd_plot.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

```
study_design_plot_2 <- ggplot() +
  geom_col(
    data = survey_ggplot,
    aes(x = fct_rev(credits_sd), y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = study_design_estimates,
    aes(x = credits_sd, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = study_design_estimates,
    aes(x = credits_sd, y = 100*probability_estimate),
    size = 2) +
  coord_flip() +
  scale_x_discrete(drop = FALSE) +
```

```
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  plot_theme_2


study_design_plot_2
```




```
ggsave(
  plot = study_design_plot_2,
  here("figures", "credits_sd_plot_2.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Data analysis credits plots

```r
data_analysis_estimates <- eff_brm.mn.dataanalysis[[1]] %>%
  clean_names() %>%
  rename(
    credits_da = cats,
    probability_estimate = estimate) %>%
  select(discipline, credits_da, probability_estimate, lower, upper)

data_analysis_estimates
```

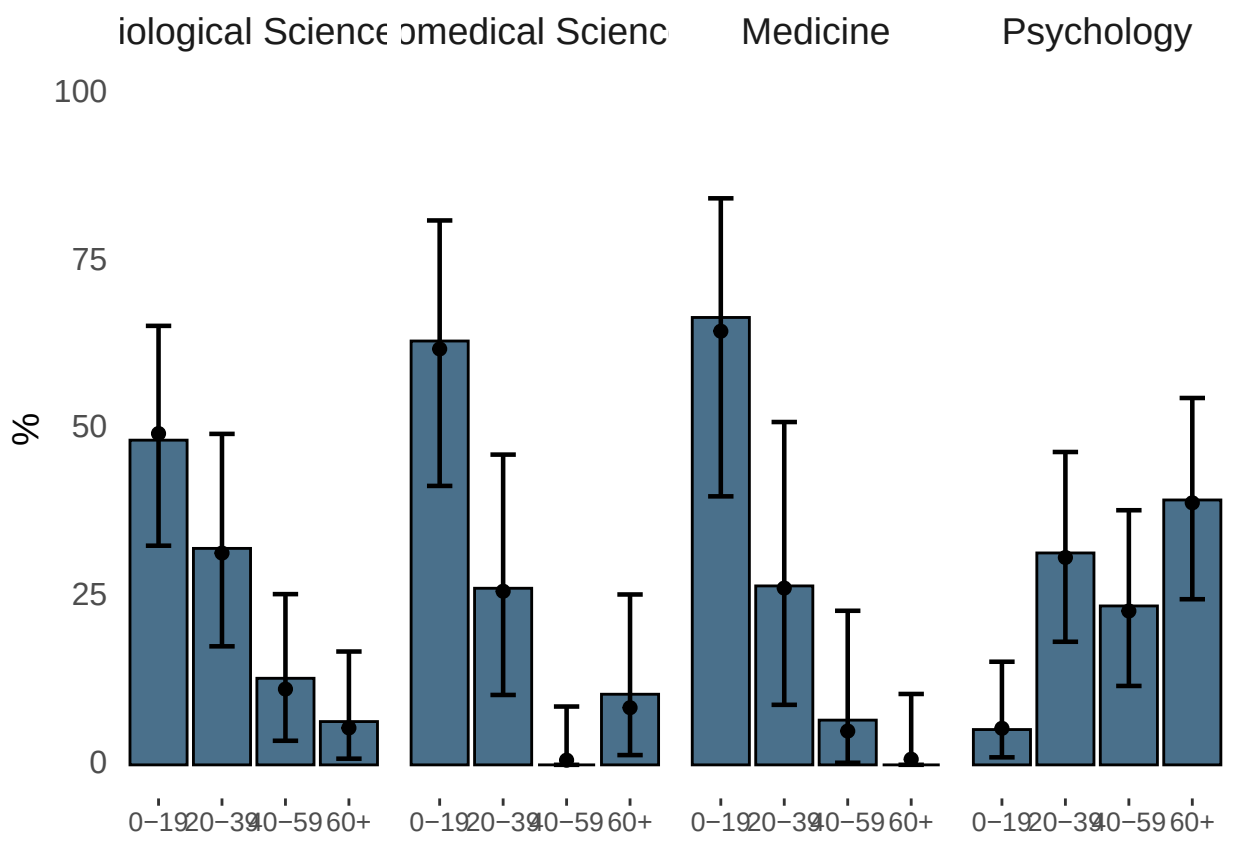
```
##    discipline credits_da probability_estimate        lower     upper
## 1         Bio       0-19          0.493711489 3.265750e-01 0.6541312
## 2      Biomed       0-19          0.619782552 4.156760e-01 0.8112096
## 3         Med       0-19          0.646203663 4.001186e-01 0.8442054
## 4       Psych       0-19          0.054446312 1.121355e-02 0.1536669
## 5         Bio      20-39          0.315587443 1.766543e-01 0.4931739
## 6      Biomed      20-39          0.258694734 1.041784e-01 0.4623235
## 7         Med      20-39          0.263437738 8.945881e-02 0.5108644
## 8       Psych      20-39          0.309021454 1.834578e-01 0.4662058
## 9         Bio      40-59          0.112930194 3.591748e-02 0.2544229
## 10     Biomed      40-59          0.007028225 6.095001e-05 0.0869425
## 11        Med      40-59          0.050358928 3.130164e-03 0.2296010
## 12      Psych      40-59          0.229262108 1.176194e-01 0.3794330
## 13        Bio        60+          0.055061795 9.194758e-03 0.1688862
## 14     Biomed        60+          0.085198453 1.457633e-02 0.2539129
## 15        Med        60+          0.008448973 8.410045e-05 0.1056258
## 16      Psych        60+          0.390477106 2.468189e-01 0.5466544
```

```r
survey_table <- table(survey_data$credits_da_recoded, survey_data$discipline)
survey_table <- prop.table(survey_table, margin = 2)
survey_table <- as.data.frame(survey_table)
survey_ggplot <- survey_table %>%
  rename(
    credits_da = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```r
data_analysis_plot <- ggplot() +
  geom_col(
    data = survey_ggplot,
    aes(x = credits_da, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = data_analysis_estimates,
    aes(x = credits_da, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = data_analysis_estimates,
    aes(x = credits_da, y = 100*probability_estimate),
    size = 2) +
  # #coord_flip() +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
```

```
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  plot_theme


data_analysis_plot
```

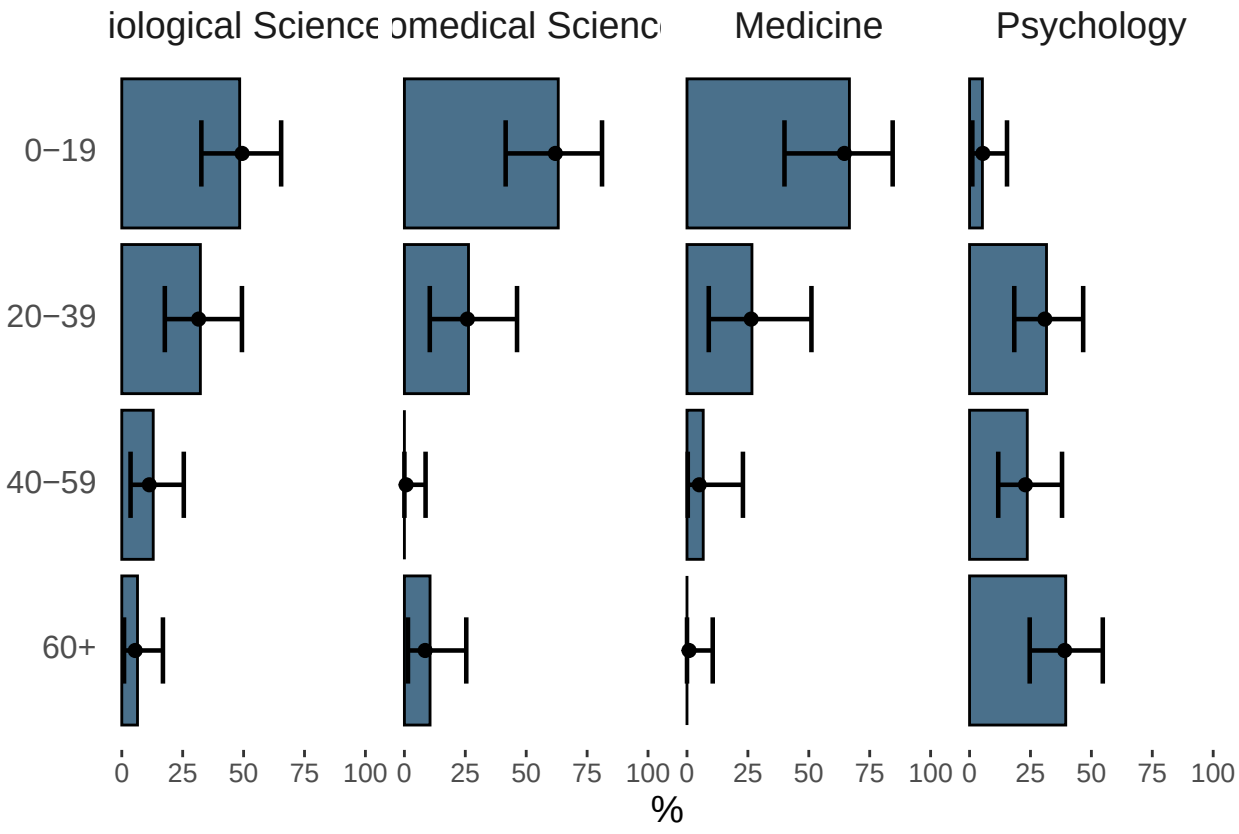



```
ggsave(
  plot = data_analysis_plot,
  here("figures", "credits_da_plot.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

```
data_analysis_plot_2 <- ggplot() +
  geom_col(
    data = survey_ggplot,
    aes(x = fct_rev(credits_da), y = percentage),
```

```
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = data_analysis_estimates,
    aes(x = credits_da, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = data_analysis_estimates,
    aes(x = credits_da, y = 100*probability_estimate),
    size = 2) +
  coord_flip() +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  plot_theme_2


data_analysis_plot_2
```



```r
ggsave(
  plot = data_analysis_plot_2,
  here("figures", "credits_da_plot_2.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```